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Burns

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SUMMARY PEARLS

- Rehabilitation after burn injury focuses on evidence-based interventions to minimize complications of burn injury and improve outcomes.
- Physiatrists' involvement with the interdisciplinary team treating burn survivors should begin at admission, continue through the inpatient rehabilitation and outpatient setting, and continue to address needs months and years after the initial injury. Physiatrists impact the long-term health and well-being of burn survivors.
- Acute management of burn injury is a critical component to survivors' short- and long-term outcomes, and an interdisciplinary team is required.
- Pain mitigation should include treatment of nociceptive and neuropathic pain, anxiety, and neuropathic itch via multimodal medications and nonpharmacologic interventions, but it remains challenging.
- Burn injury results in long-lasting widespread systemic effects to include metabolic rate alterations and unique nutritional needs, as well as local inflammatory and signaling pathway changes that affect the healing process and can lead to development of hypertrophic scarring.
- The physiatrist diagnoses and treats postburn complications, including pruritis, neuropathy, heterotopic ossification, care of amputated limbs' pain, nutrition, dysphagia, cognitive dysfunction, psychological coping and mood, and community reintegration.
- The physiatrist devises and oversees an exercise program based on the survivor's functional status and goals, to include aerobic and resistive exercises, cardiovascular conditioning, and work hardening.
- Psychological consequences of burn injury can include anxiety, depression, sleep disturbances, difficulty with body image and self-esteem, and changes in sexuality and sexual functioning, and these issues should be addressed.
- Community reintegration is a key component of burn survivors' recovery.

Introduction

Burn rehabilitation focuses on a comprehensive approach to survivorship, utilizing a multidisciplinary team of specialists to focus on the physical, functional, and psychological recovery of those surviving burn injuries. Burn rehabilitation is specialized to each survivor's needs to achieve personalized outcomes and goals. Physiatrists, given their unique skill set, can and should follow burn survivors for months to years after the initial injury.

As mortality after burn injury continues to fall (with a nearly 98% survivorship rate),⁷ focus is shifting to evidence-based strategies for long-term prevention of complications, optimization of rehabilitation, and interventions to promote psychological wellness and social recovery. To achieve these goals, the National Institute on Disability and Rehabilitative Research has created and funded the Burn Model Systems database.⁹³ This resource, along with the American Burn Association's (ABA) repository, has allowed data-driven advances in care of individuals throughout their lifetime.

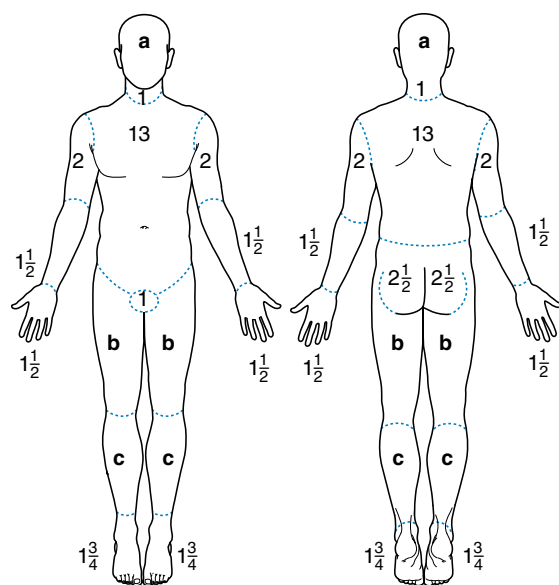
Classification

Burn injury results in tissue damage caused by thermal (flame), chemical, electrical, or radiation exposure. Burn injury severity

and subsequent treatment are defined by the size and depth of the burn, which should be documented upon initial presentation to a burn center but may evolve as the burn wound develops.

Burn Size

The size of the burn is defined by the percentage of total body surface area (TBSA) involved. Generating a TBSA percentage is the basis for fluid resuscitation, transfer decisions, and prognosis of complications and mortality. This is typically estimated using the Lund-Browder chart (Fig. 27.1), which approximates the percentage of body area in each limb, the head, torso, and genitalia. Utilizing different percentages based on relative size of body parts for young children compared to adults improves accuracy of TBSA calculation. Each area of burn can be further broken down based on the percentage of each area affected. For example, if half of the anterior right arm is burned, it can be estimated to involve 0.75% TBSA. Smartphone applications, digital planimetry, and laser Doppler are also used in some centers.³¹ The Lund-Browder chart is most helpful for cutaneous burns from high thermal sources such as flame. It will not give an accurate representation of damage below the surface of the skin, such as that present from electrical injury.



	Age in years					
	0	1	5	10	15	Adult
a - 1/2 of head	9 1/2	8 1/2	6 1/2	5 1/2	4 1/2	3 1/2
b - 1/2 of one thigh	2 3/4	3 1/4	4	4 1/4	4 1/2	4 3/4
c - 1/2 of one leg	2 1/2	2 1/2	2 3/4	3	3 1/4	3 1/2

• **Fig. 27.1** The Lund-Browder method for calculation of burn size is reliable for different body proportions in children. (From Artz CP, Moncrief JA, Pruitt BA. *Burns: A Team Approach*. Saunders; 1979.)

Burn Depth

Burn depth is the other measure of burn severity. Burns are classified into superficial, superficial partial thickness, deep partial thickness, and full thickness depending on the layers of skin involved (Figs. 27.2 and 27.3). Each classification has unique characteristics, but often multiple depths of burn can be seen in one injury.

Superficial burns involve only the epidermis. The function of the epidermis is to provide a barrier from external pathogens and prevent moisture loss.⁷⁰ Superficial burns are red, warm,

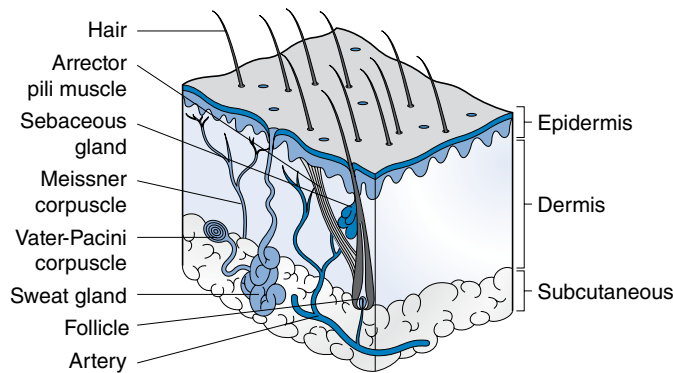
painful, and without exudate. They typically heal within 7 days, without intervention such as dressings or surgical debridement, as the epidermis regenerates from the basal cell layer. Superficial burns are not included as part of the Lund-Browder calculation of burn size.

Partial-thickness burns produce damage to the level of the dermis. The dermis is primarily composed of collagen and protein; it contains the dermal appendages such as hair follicles, exocrine sebaceous and sweat glands, cutaneous nerves (transmitting touch, vibration, proprioception, and pain), and arterial and venous capillaries.⁷⁰ Superficial partial-thickness burns damage the upper levels of the dermis. Due to loss of the epidermis, fluid may be contained within a blister, or if the blister is unroofed, leak. Partial-thickness burns weep and are painful due to preserved nerves within the lower portions of the dermis. Additionally, capillaries remain intact, so these burns blanch with pressure. Superficial partial-thickness burns may be able to reepithelize via basal cell migration if remnants of hair follicles and other adnexa that contain basal cells remain intact.⁸⁶

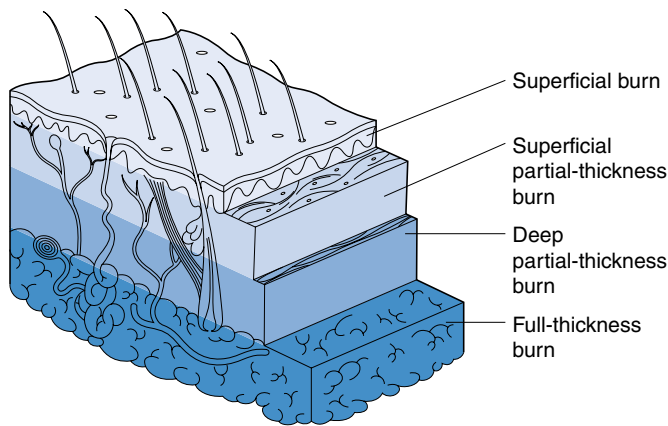
Deep partial-thickness burns affect the deeper layers of the dermis and typically do not reepithelize well due to damage to most of the remaining basal cell layers and persistent inflammatory state.^{70,86} Because of these factors, attempted healing of deep partial-thickness burns is prone to development of dysfunctional skin with hypertrophic scarring or contracture.⁴⁰ Wound care is initiated, but these burns typically require debridement, placement of allograft or dermal substitutes, and eventual autograft for the best functional and cosmetic outcome. Deep partial-thickness burns are extremely painful.⁷⁰

Burns that breach the basement membrane of the dermis and expose subcutaneous fat, muscle, or bone are considered full thickness.⁷⁰ These burns do not blanch or produce exudate due to the destruction of capillaries. They may be white and waxy in appearance or dark from necrotic coagulated tissues. If deeper subcutaneous structures such as major blood vessels are involved, limbs may be nonsalvageable and require amputation. Full-thickness burns are less painful than partial-thickness burns given destruction of cutaneous nerve fibers.⁷⁰ Standard of care is prompt excision of eschar to reduce nidus of infection, reduce inflammatory response attributed to necrotic tissue, and reduce potential for compartment syndrome.^{86,141}

Other injuries requiring the specialized care at a burn center include acute frostbite, Stevens-Johnson syndrome, toxic epidermal necrosis, and necrotizing fasciitis.⁵¹



• **Fig. 27.2** Cross section of normal skin, illustrating the epidermis, the dermis, and the epidermal appendages (hair follicles, sweat, and sebaceous glands). (From Achauer BM, Eriksson E. *Plastic Surgery Indications, Operations and Outcomes*. Mosby; 2000.)



• **Fig. 27.3** A superficial (first-degree) burn involves only the epidermis. A partial-thickness (second-degree) burn involves the superficial or deep dermis, and a full-thickness (third-degree) burn extends to subdermal tissues. (From Achauer BM, Eriksson E. *Plastic Surgery Indications, Operations and Outcomes*. Mosby; 2000.)

• BOX 27.1 Criteria for Transfer to a Certified Burn Center

- Any burn involving the face, hands, feet, or genitals
- Inhalation injury
- Greater than 20% partial-thickness burn in those older than 10 years and younger than 50 years
- Greater than 10% partial-thickness burn in those younger than 10 years and older than 50 years
- Any full-thickness burn >5% total body surface area
- Electrical burn
- Chemical burn

Initial Resuscitation

In addition to defining burn size and depth, acute care is initiated by the receiving team per the Advanced Burn Life Support (ABLS) protocol.⁶ ABLS includes assessment of airway (to determine presence of inhalation injury and/or need for intubation), breathing, and circulation, with emphasis on fluid resuscitation utilizing established protocols. Secondary survey for additional traumatic injuries, carbon dioxide levels, and pain control are also addressed.

Injury characteristics requiring transfer to a certified burn center for specialized care and resuscitation are outlined in [Box 27.1](#).⁷

Electrical Burn

Electrical injury is a special consideration requiring immediate transfer to a certified burn center because these injuries are more likely to result in neuropathy, burn-related fasciotomies, amputations, and longer length of hospital stay compared to thermal injury.^{72,159} Electrical burns are arbitrarily divided into those due to high voltage (>1000 V), often encountered in a workplace setting, and low voltage (<1000 V), likely to occur at home. Electrical injury typically accounts for approximately 5% of all admissions to a burn center.¹⁶⁰ Typically, there are relatively small cutaneous entry and exit points, but there may be significant damage to underlying structures. Central and peripheral nerves are particularly susceptible because nerves, by their nature, are

electrical conductors.⁹ Clonus may be seen as a later-onset complication of electric damage to central and peripheral nerves.⁴⁴

Muscle is also preferentially affected by electrical injury. Damage to the cellular lipid bilayer of myocytes leads to progressive necrosis and tissue edema, resulting in compression and compartment syndrome.¹⁶⁰ Additional consequences of muscle and myocardial necrosis include rhabdomyolysis and renal failure via myoglobinuria.¹⁶⁰ Damage to blood vessels can result in avascular necrosis of limbs or end organs.¹⁵⁹ Electrical injuries can also cause damage to ocular structures (retina and lens); thus ophthalmology evaluation is required.¹⁶⁰

Concomitant Diagnoses

Inhalation Injury

The presence of inhalation injury (along with burn size and age) is one of the major predictors of mortality and should be assessed promptly upon presentation. In those requiring hospitalization for severity of the thermal burns, reported incidence of inhalational injury varies from 5%¹⁷⁹ to 20% to 30%.¹⁵² The risk of death in the presence of inhalation injury is approximately 30%, compared to 2% in survivors with cutaneous burns alone.¹⁶³ In addition, there is a higher likelihood for development of complications such as pneumonia, adult respiratory distress syndrome, and multisystem organ failure where inhalation injury is present.¹⁵² Awareness of inhalation injury at the time of resuscitation is helpful to predict decreased pulmonary diffusion capacity and respiratory muscle weakness leading to decreased exercise tolerance during rehabilitation.¹⁸⁶ Inhalation injury is also an independent risk factor for dysphagia.³⁶

Fractures

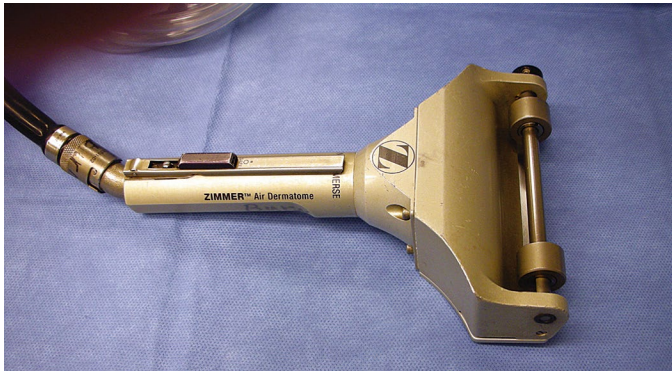
Traumatic fractures are reported to occur in 5% to 7% of burn survivors presenting to burn centers and should not be overlooked on survey especially in those with burn injury from car crashes.⁴⁸ Coordination of care between services is essential to reduce infection risk, ensure proper timing for fixation of fractures, and determine progression of weight-bearing status that may influence postacute rehabilitation.

Brain Injury

Concomitant brain injuries from direct trauma, blast effect, or anoxia can also complicate acute care management and rehabilitation. Resultant agitation or confusion due to brain injury may impede survivors' understanding and cooperation with wound care, range of motion (ROM), weight-bearing status, adherence with medical devices, and positioning.¹⁷¹ The first-line intervention is always standard delirium precautions. If restraints are needed for safety, care should be taken to avoid placing over burned or grafted areas. Standard medications for agitation can be employed, such as propranolol and quetiapine.

Surgical Management

Excision and debridement within 48 hours of burn injury has been found to reduce complications, intensive care unit (ICU) length of stay, and hospital length of stay.¹⁴¹ Full-thickness, circumferential burns to the trunk or extremities are prone to tissue swelling resulting in compartment syndrome, requiring prompt



• **Fig. 27.4** Donor skin is removed from an unburned area of skin through the use of a powered dermatome.

fasciotomy. Excision and preparation of the wound bed may entail multiple trips to the operating room throughout the survivor's hospital stay depending on burn size and medical stability. Allograft from donated cadaveric skin or biologic synthetic dressings can be applied to protect the wound bed and minimize insensible losses.⁷⁴ Once the survivor is hemodynamically stable and the wound bed is clean and viable, autografting is pursued. Typically, split-thickness skin grafts (STSG) are harvested from an unaffected area of the survivors' body. A dermatome (Fig. 27.4) will remove the epidermis and a portion of the dermis. The deeper portions of the dermis at the donor site will provide reepithelialization of donor sites within a few weeks. The donor site is typically extremely painful until reepithelialization occurs. The STSG is then meshed to increase the surface area coverage. The interstices between the meshed areas close by secondary intention. If deemed necessary, closure time can be augmented with epidermal autografts (ReCell, Valencia, CA) to accelerate healing.⁷⁹ In cosmetically sensitive areas, such as the hands, feet, and face, sheet grafts (unmeshed STSG) are used to minimize the meshed appearance of scarring.⁸⁷

In areas of circumferential burn to limb or torso, tissue damage and inflammatory swelling can lead to increased volume within a space surrounded by uncompressible fascial planes, leading to further tissue ischemia and damage referred to as compartment syndrome.¹⁰⁴ Compartment syndrome is diagnosed clinically by assessing pulses, palpation to evaluate for tense compartments, and increased pain on passive ROM.¹⁰⁴ Compartment syndrome can lead to perfusion deficit and potentially further damage to distal tissue and may require fasciotomy to correct.⁷⁴ If the damage to the tissue from the burn injury itself or compartment syndrome is beyond salvage, an amputation may be considered.

Physiatric Evaluation and Functional Assessment

Early physical medicine and rehabilitation consult by a physiatrist allows involvement throughout the patient's acute care course to make appropriate rehabilitation-specific recommendations and evaluate the patient's candidacy for inpatient rehabilitation (IPR). Utilization of physical (PT) and occupational therapy (OT) services during the acute care phase is standard of care.

When evaluating a burn survivor in the acute care setting, the rehabilitation team should document pertinent portions of the history and physical exam to inform the treatment plan. Essential

elements of the history include how the burn occurred, etiology, classification and TBSA of the burn injury (including the initial modified Lund-Browder chart documented by the acute care team), concomitant injuries, plans for surgical salvage vs. amputation of affected limbs, weight-bearing status, additional planned interventions or surgeries, pain and current management strategy, need for perioperative positioning and splinting, preinjury functional status and cognition and activity level, current ability to perform activities of daily living (ADL) and mobility, psychosocial concerns, and short- and long-term treatment goals. The exam should include measurement of edema, ROM, strength, sensation, and scar assessment.⁷⁶

Treatment paradigms are shifting away from postgraft bedrest toward early ambulation to minimize effects of immobility.⁵⁸ Practice guidelines established in 2012 advocated for early ambulation (within ≤ 1 day of graft application) for those with isolated lower extremity skin grafts of relatively small TBSA.¹¹³ A soft compressive dressing should be in place, and grafts crossing joints should be splinted into neutral.¹¹³ Ambulation on the day of lower extremity grafting (with a soft compressive dressing in place) did not result in graft loss, shear, or hematoma formation, even if grafts crossed a joint.⁵⁸ Several studies have also demonstrated feasibility of mobilizing burn survivors with larger TBSA and those requiring mechanical ventilation with improvement in ICU and hospital length of stay.^{43,115} Data collected during early ambulation, such as level of assistance needed, endurance, pain with mobility, and joint ROM during ambulation, can help inform the needed rehabilitation resources at the time of acute care discharge.

Pain Management After Burn

Pain after burn injury is often significantly experienced regardless of burn depth and size. Adequate analgesia requires a multimodal approach to reduce barriers to treatment and rehabilitation.⁶³ A generalized understanding of pain pathophysiology and management is imperative to treating burn survivors.¹⁵⁰ An adequate pain management regimen should address all facets of pain, as follows:

- *Background pain* is present at rest and constant, due to nociceptive activation as a direct result of tissue damage, pressure, and the inflammatory milieu.^{84,92,148,182}
- *Episodic pain* is described as an escalation of pain without an inciting stimulus and is of relatively short duration.¹⁸²
- *Procedurally evoked pain* is typically of higher intensity, although reduces when the precipitating intervention (dressing changes, manipulation and debridement of wounds, aggressive stretching with therapy, operative interventions) ceases.⁸⁴ Procedural pain can be encountered multiple times per day.
- *Donor site skin harvest* produces new areas of exquisite pain.¹⁴⁸
- *Anticipatory anxiety* also requires treatment because it can heighten the pain experience.⁸⁴

Adequate treatment of acute pain is imperative to reduce central sensitization that can progress to chronic pain and to facilitate participation in rehabilitation and recovery.^{63,84,182} In the acute phase of burn treatment, opioids are the most effective means to mitigate moderate to severe pain.¹⁴⁸ Opioid selection and route of administration should consider individuals' medical comorbidities, current level of respiratory support, and physiology¹⁴⁸; however, preference of agent tends to be center specific because no consensus or clinical trial data provide recommendations for one opiate over another in the burn population.¹¹⁰ In cases of a burn survivor requiring intubation and sedation for physiologic support in the ICU or intraoperatively, continuous infusion of

opioid pain medications may be appropriate.⁸⁵ After extubation, intravenous administration via patient-controlled analgesia (PCA) pumps or intravenous push by a nurse can be used, with the goal to transition to an oral formulation for discharge to home or the next level of care. For those with hand burns, PCA controls may need special modifications.

While the goal is always preferential downtitration of opioids over other pain modulators,¹⁴⁸ it can be expected that those with a protracted course will require a high dose (and at times higher than recommended dose)¹⁸ of opioid analgesia accounting for tolerance after prolonged exposure as well as physiologic differences in metabolism of opioids⁴⁹ after burn injury. Well-established complications of opioid treatment in the burn population are similar to nonburned patients, including tolerance, nausea, constipation, pruritis, and respiratory depression. The amount of opioids required by burn survivors directly correlates with longer-term negative outcomes such as opioid dependence or abuse, mental and behavioral disorders, and self-harm in this population.¹⁸⁰ In one retrospective study, 4.2% of opioid-naïve burn survivors met criteria for long-term opiate use or opiate use disorder 1 year after injury,²⁴ further underscoring the need for adequate acute pain regimen with discontinuation of opioids as soon as possible.

Opioid pain medications should never be administered in isolation given the elevated risk of harm in this population.¹⁴⁷ Instead, multimodal analgesia integrates nonpharmacologic and pharmacologic interventions to target diverse mechanisms of pain, which synergistically provide elevated pain relief compared with one intervention alone.³³ Burn survivors who receive multimodal analgesia have less opiate medication requirements while inpatient without increase in pain levels⁴⁷ as well as at discharge and at subsequent follow-up,¹⁸⁸ reducing potential for tolerance and dependence.

Adjuncts to manage acute nociceptive pain in burn survivors include ketamine and buprenorphine, although this varies by institution.⁸⁴ Continuous nerve block infusions utilizing bupivacaine have been shown to reduce the usage of opioid pain medications after donor site harvest⁷⁸ and have high-quality evidence for superior efficacy to opioids.³³ Although not specific for pain after burn injury, coadministration of acetaminophen or nonsteroidal antiinflammatory drugs (NSAIDs) with opiates is strongly recommended based on high-quality evidence.^{33,148}

Clinicians should also be aware of opioid-induced hyperalgesia (OIH), in which an increase in dosage of opiates results in higher levels of pain instead of amelioration. Proposed mechanisms include a complex interplay between afferent nerves, spinal cord dorsal horn neurons, spinal glutamate transporters, circulating cytokines, and modulation at the rostral ventromedial medulla.⁸⁰ OIH may be difficult to differentiate from tolerance in burn survivors. Pain in OIH is described as widespread allodynia and hyperalgesia, even outside of the original area of pain.⁸⁰ This pain worsens with increased dosages of opioids. A difficult clinical decision to reduce the amount of opioid and assess carefully whether pain improves can help differentiate tolerance from OIH.⁸⁰ Additional strategies of opioid rotation and adjunct nonopioid medications may be helpful. Preclinical research is ongoing to refine opiates to preferentially activate ligands of the mu, kappa, and delta nociceptive receptors that reduce transmission of pain signals and avoid those ligands that facilitate undesired effects.⁷⁷

Neuropathic Pain

Neuropathic components of burn-induced pain are expected given extensive damage to cutaneous nociceptors and nerve fibers.

Contributors include pathologic firing of peripheral nerves as well as amplification of pain signals due to peripheral and central sensitization.¹¹⁰ Neuropathic pain after burn injury is similar in description to neuropathic pain due to conditions such as diabetic and postherpetic neuralgia.¹⁵⁵ However, limited and at times conflicting data exist on the effects of neuromodulating medications such as gabapentin and pregabalin,^{92,183} which are effective in nonburn neuropathic pain. In both retrospective¹⁸⁷ and prospective studies,⁸⁵ pregabalin in dosages up to 300 mg twice daily decreased the pain score in burn survivors and was well tolerated. Gabapentin dosed prior to surgical debridement of burn wounds demonstrates improvement in pain levels and reduced opioid requirements postprocedure.¹⁴⁴ Decreased opioid requirements when gabapentin is utilized as part of a standard pain protocol have been reported.^{68,94} However, one small randomized and placebo-controlled study found no difference in incidence of neuropathic pain scores, postprocedural pain scores, opioid requirements, or functional status both during hospital stay and at the time of first posthospital follow-up with the addition of neuropathic pain medications.¹⁸³ Despite the paucity and at times conflicting evidence, the ABA recommendations include consideration of neuropathic pain-modulating agents in those with neuropathic pain features or who are refractory to standard therapy.¹⁴⁸

Nonpharmacologic Pain Management Techniques

The ABA strongly recommends augmenting medical pain management with nonpharmacologic pain relief.¹⁴⁸ This can include cognitive behavioral therapy (CBT) to reinterpret and reframe the pain experience, virtual reality to distract from current pain level by immersing into a novel environment,¹⁴ and relaxation techniques such as deep breathing or listening to music to decrease anticipatory anxiety that perpetuates pain.¹⁰⁸ Given discrepancies between hospital resources and methods involved in study design, there is no recommendation of one nonpharmacologic intervention over another. However, nonpharmacologic intervention has not been associated with harm to burn survivors and thus is a strongly recommended adjunct for pain management.¹⁴⁸

Complications of Burn Injury

Burn injury results in a wide range of local and systemic complications. It is important for the physiatrist to recognize and treat these issues that will impact outcomes and function if not addressed. In addition, these complications may present acutely or manifest months to years later in recovery, further underscoring the importance of long-term physiatric assessment.

Hypermetabolic State and Nutritional Support Needs

Nutrition presents a challenge within burn rehabilitation. The resting energy expenditure of a survivor of severe burn injury is significantly increased (40% to 100% more than baseline)⁷³ due to circulating catecholamines and their effect on adenosine triphosphate-driven metabolic processes, including protein synthesis, hepatic gluconeogenesis, and urea production, among others.¹⁹¹ Increased mitochondrial oxygen consumption for thermogenesis to compensate for heat lost through skin disruption and thus maintain body temperature also plays a role.¹³³ Skeletal muscle catabolism outpaces skeletal muscle synthesis, leading to decreased lean muscle mass, which may persist for years.¹³³ Optimizing nutrition includes early enteral feeding to offset the massive increase in metabolic rate that can last for 1 to 3 years.¹¹⁴

Nutritional support is therefore critical to recovery and rehabilitation acutely and chronically after burn. Total caloric needs

are anywhere between 20% and 60% greater than baseline.¹⁸⁵ Multiple formulas have been used to determine caloric needs after burn injury and guide supplementation, although no consensus yet exists.¹¹⁴ Care should be taken not to overfeed when metabolic needs wane.

Current recommendations to decrease skeletal muscle breakdown during the hypermetabolic state of burn recovery include a diet that is high carbohydrate (55% of calories), moderate protein (20% to 25% of calories), and low fat (<20% of calories).¹⁶⁴ High-carbohydrate components are recommended as glucose is a preferred energy source, sparing breakdown of intrinsic protein that can be utilized for wound healing and reducing muscle mass breakdown.¹¹⁴ Optimal protein intake facilitates protein synthesis and reduces nitrogen loss.¹²⁵ In addition, supplementation of vitamins and trace minerals, in particular A, E, C, D, zinc, and folic acid, as deficiencies are shown to lead to impaired immune function and wound healing with increased morbidity.¹⁶⁴

In addition to increasing appropriately balanced caloric intake, several pharmacologic interventions may be used to mitigate the effects of the hypermetabolic state. Oxandrolone, an analog of testosterone, is typically dosed 10 mg twice daily and may be used for up to 2 years to promote protein synthesis and reduce muscle protein loss.⁴² Important benefits of oxandrolone include improving lean muscle mass, protein synthesis, bone density, and muscle strength and reducing time between surgeries. The 1% risk of hepatic dysfunction and hirsutism should be kept in mind, but overall, oxandrolone poses minimal risk. Despite these benefits, oxandrolone has now been removed from the US market but may be available in other countries.¹⁸⁵ Human growth hormone shows similar effects on burn wound healing compared to oxandrolone, but the significant expense, hyperglycemic effects, and increase in metabolic rate have led it to fall out of favor.^{22,41} Propranolol reduces cardiac work, liver steatosis, and muscle catabolism through beta-adrenergic blockade and may eventually become a standard of care for preserving muscle mass after burn injury.¹²⁸ Metformin or insulin may be used for hyperglycemia and insulin resistance that occurs as part of the stress response to burn and result in poorer outcomes. The suggested goal for postburn glucose levels is 130 to 140 mg/dL.² In many cases, the combination of these pharmacologic interventions provides positive outcomes compared to controls.¹⁸⁵

Dysphagia

Dysphagia of varying degrees was detected in 27.78% of people surviving a major burn.¹²⁸ Dysphagia after burns can complicate the provision of adequate nutrition intake and place survivors at risk for additional complications such as aspiration pneumonia, dehydration, or insufficient intake to meet their postburn nutritional needs. Risk factors for dysphagia after burns include larger TBSA (>18%), older age, head and face involvement, presence of inhalation injury, need for prolonged mechanical ventilation, and need for tracheostomy or escharotomy.¹²⁸ Generalized muscle weakness and catabolism may also contribute to development.¹⁵¹ All burn survivors should have an initial screening for dysphagia, especially if they have had a protracted course including ventilation, immobilization, or prolonged nonoral means of nutrition prior to diet initiation.

Objective evaluation of swallowing function can be completed by functional endoscopic evaluation of swallow (FEES) or modified barium swallow (MBS). For those with identified dysphagia, visits five to seven times per week with a trained speech-language pathologist (SLP) are recommended. Treatment includes standard

oropharyngeal swallowing exercises, Frazier free water protocol, and trials of advanced textures in a controlled environment before diet upgrades.⁶¹ Some patients will need multiple evaluations to assess progress. Improvement in swallowing function of those with dysphagia is the most rapid between weeks 2 and 6 after injury.¹⁵¹ Of patients with initial dysphagia, 85% had complete resolution of dysphagia within 10 weeks after injury.¹⁵¹ Therefore 15% of patients may have prolonged dysphagia after discharge from the hospital setting or IPR. Continued dysphagia treatment in the outpatient setting, including additional objective evaluations (MBS or FEES), is recommended.

Cognitive Concerns

All burn survivors should be given a cognitive screen and, if it is positive, receive continued cognitive therapy by an SLP. Up to 79% of burn survivors have cognitive communication deficits on admission to IPR.^{75,138} Burn patients have lower memory scores on measures of cognitive function compared to other populations in rehabilitation.¹³⁸ This holds true after accounting for confounders. The most common etiology is delirium, but other etiologies may include decreased oxygenation to the hippocampi from smoke or chemical inhalation, stress of the injury and hospital setting, and medications.¹³⁸ For those with deficits, speech-language therapy is initiated and typically dosed 5 to 7 days per week in 30- to 60-minute sessions. The treatment focuses on maintaining sustained attention, memory strategies, functional problem solving, and carryover of techniques and treatments. Review of medications and their purpose upon discharge to home is also addressed to ensure adherence. The frequency of SLP sessions can be decreased as cognitive status improves. However, in 27% of survivors, deficits can persist past discharge (similar to data described in post-ICU survivors).⁷⁵ Outpatient referral for cognitive SLP service is recommended for those still with deficits at the time of discharge.

For burn survivors discharged to IPR, a higher cognitive score on admission is associated with a higher functional score at discharge. Patients with higher cognitive scores on admission to IPR are also less likely to be discharged back to acute care within the first 72 hours of their IPR stay.¹¹

Postburn Pruritus

Pruritis often accompanies neuropathic pain in burn survivors.⁸⁸ Pruritis affects between 70% and 90% of adult burn survivors and reduces quality of life.^{26,157} Intrinsic factors that may confer greater itch severity postburn include female sex, older age, a larger TBSA, and those with burns located on the head/face/neck.¹³⁴ Although no consensus has been reached, pathophysiology of postburn pruritis is postulated to include pathways involved in regeneration of tissue¹⁵⁷ as well as a separate and specialized pathway to transmit neuropathic itch.⁸⁸ Pruritis sensation is experienced due to the activation of cutaneous itch receptors by substances present in the burn scar, such as histamine, opioids, tumor necrosis factor-alpha, cytokines, proteases, and acetylcholine. These neuropeptides are needed to promote epidermal, connective tissue, and vascular cell proliferation; however, they also increase sensitivity of the pruriceptors.^{34,157} Sensation from the pruriceptors is transmitted by afferent C-fibers.¹²⁷ The signal is relayed to the dorsal root ganglia and then the dorsal horn of the spinal cord, crossing to the contralateral side and further conducted centrally by the spinothalamic tract.¹²⁷

Gabapentin has been shown to reduce pruritis scores in burn survivors,⁶⁴ as has a separate study on pregabalin,⁵ suggesting a

neuropathic itch travels via a similar mechanism of transmission to neuropathic pain postburn. This is further supported by evidence that antihistamines such as cetirizine offer no additional benefit for postburn pruritis in burn survivors.^{5,34} A topical compound cream containing 5% to 10% ketamine, 5% amitriptyline, and 5% lidocaine has also shown improvement in postburn neuropathic itch.⁹⁶ Topical emollients and massage also have a role in treatment. Adjunct treatments with botulinum toxin, naltrexone, tricyclic antidepressants, selective serotonin reuptake inhibitors, and behavioral therapies have also been described but are not widely used.^{34,157}

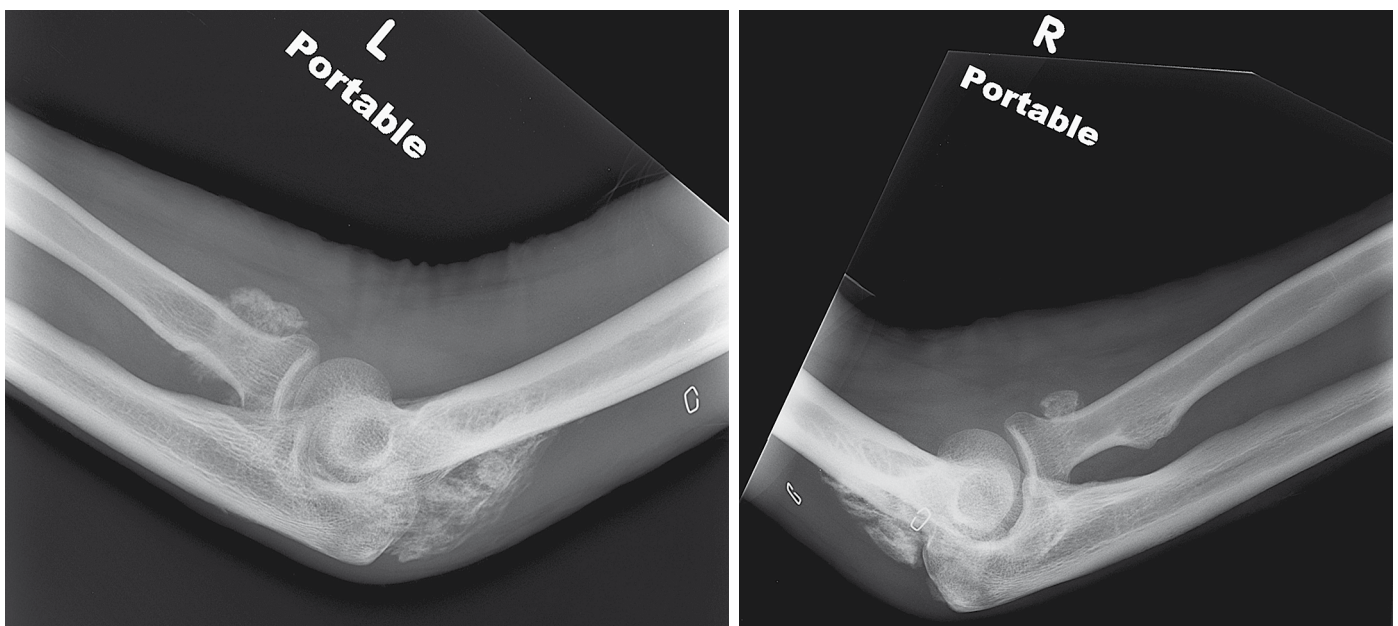
Neuropathy

Between 10% and 37% of burn survivors are affected by peripheral neuropathies.^{34,142} Diagnosis of focal and generalized peripheral neuropathy is made with careful neurologic examination and/or electrodiagnostic testing.⁵⁵ As in the general population, neuropathies may be managed in IPR setting with PT, OT, splinting, and neuropathic pain medication.¹⁷⁴ Median mononeuropathy is the most common mononeuropathy in this population.¹⁷⁴ Causes of mononeuropathy include direct thermal or electrical injury, damage related to escharotomy or fasciotomy, soft tissue edema, metabolic derangements, immobilization, and heterotopic ossification (HO).¹⁷⁴ Additional risk factors can include preexisting injuries, older age, history of alcoholism, and diabetes mellitus.^{157,174} Generalized peripheral neuropathy, similar to critical illness neuropathy, can affect 3% to 30% of patients due in part to circulating inflammatory mediators from the body's systemic response to burn injury leading to nerve injury. This manifests clinically as paresthesia and distal weakness.¹⁵⁷ To minimize iatrogenic contribution to neuropathy, care should be taken during surgical and postoperative positioning to reduce traction on the brachial plexus resulting in plexopathy. Pneumatic inflation to minimize surgical blood loss should be carefully positioned and timed to reduce the risk of compression neuropraxia. Restraints should be padded and discontinued as soon as possible.

Heterotopic Ossification

HO is defined as mature bone deposited in areas of soft tissue, including muscle, ligaments, and tendon.¹⁴² HO affects between 3.5% and 5.6% of persons with burns,^{98,117} with the posterior elbow being the most common site (Fig. 27.5).⁹⁸ It is possible that smaller areas of HO may not cause pain or functional limitations and are therefore not diagnosed. The diagnosis of HO can be suspected clinically by pain and restricted ROM around the affected joint, a palpable firm lump, and swelling or erythema. It should be noted that HO will likely be evident clinically before corresponding changes on x-ray or triple phase bone scan. Ultrasound may also be helpful for diagnosis.¹⁰⁰ In one study, the median time to clinical development of HO in burn survivors was 37 days (range 30 to 40 days), and the median time to radiologic development was 49 days (range 38 to 118 days).¹¹⁷ Risk factors for development of HO include greater than 30% TBSA burn and arm burns requiring grafting.¹⁰⁵ The likelihood of HO also increases with each return trip to the operating room and each day requiring mechanical ventilation.¹⁰⁵

Articles since the 1970s have documented progressive ROM exercises to reduce the occurrence and prevent the progression of HO in patients with a variety of diagnoses (joint replacement, spinal cord injury, brain injury, etc.).¹⁶⁷ However, a small study specific to burn injury by Crawford et al. showed active ROM within the pain-free arc was helpful in reducing the progression of HO. Those who were engaged only in passive ROM showed progression to bone maturity and need for surgical removal.³⁸ In retrospect, this may be confounded by the severity of injury, ability of patient participation due to sedation, and length of required ventilation. Therefore in IPR, active progressive ROM guided by therapists is recommended with gentle prolonged stretch near the end range. While there is no consensus on treatment, NSAIDs have been shown to help reduce the progression of HO in burn patients but are not helpful prophylactically.¹⁰⁵ Bisphosphonates and focused radiation treatment have not shown consistent benefit and can have long-term safety concerns.¹⁰⁵ Surgical resection



• Fig. 27.5 Heterotopic ossification of the posterior elbow.

of HO can be completed to assist with ROM restoration. Timing of resection can vary from 5 to 18 months post burn; however, early resection (mean of 5.35 ± 1.46 months after burn)¹⁰⁵ with an appropriate postoperative therapy program demonstrated extensive improvement in ROM without additional complications.⁹⁸

Amputation

The incidence of reported amputation as a result of burn injury is between 2%⁷¹ and 6%.¹⁶⁵ While fire/flame remains the most likely etiology of burns requiring amputations, electrical injury has a higher incidence of amputation.¹⁶⁵ Other contributing factors to amputation include male sex, age range 18 to 64 years, higher TBSA, presence of infection, and full-thickness injury.^{13,165} The majority (~65%) of postburn amputations are considered minor (to include fingers or toes), and the remaining 35% are considered major (to include transradial, transhumeral, transtibial, or transfemoral levels).¹³

Amputation is classified as early when it is within 2 to 3 days of admission and is usually undertaken due to obvious tissue necrosis, nonsalvageable trauma such as vascular disruption or fracture, or infection. Early amputation is associated with decreased infection, mortality rates, and hospital length of stay for burn patients.¹⁹⁰ Amputation may be delayed/late (postburn day 3 and on) to determine amount of salvageable tissue remaining or feasibility of reconstruction.¹⁹⁰ If adequate healthy tissue is available, a free-flap tissue transfer may reduce the risk of future amputation.⁸⁹

Timing of prosthetic training and rehabilitation in the burn survivor is similar to amputees without burn injury. It includes complete closure of all incisions and initiation of edema management and shaping systems, such as compression socks or dressings or rigid removable orthoses.⁵⁴ Prosthetic fit after burn-related amputation presents unique challenges. The presence of amputation is a major contributor to contracture formation in burn survivors.¹⁵⁶ Other considerations for postburn amputees include the likeliness of adherent skin tissue due to reduced elasticity of grafted skin, reduced sensation or hypersensitivity, and reduced ability to perform desensitization due to grafted skin or ROM limitations.⁵⁴ Burn-related heterotopic ossification can also be a confounding factor in prosthetic fit.⁶³ The prosthetic prescription should account for these parameters, as well as viability of skin

grafts or flaps (including vascular anastomoses), potential repeated skin breakdown, and concomitant joint contractures due to scar or positioning.^{89,135} All amputees should engage in a progressive wear schedule to ensure tolerance over burned skin and grafts to minimize shear and risk of skin breakdown.⁵⁴

Socket design should ensure pressure is in load-tolerant areas. Total surface bearing sockets are preferred to patella tendon bearing sockets in burn-related transtibial amputees to increase the load bearing surface.⁵⁴ If required for proper fit, the liner interface can be custom molded to accommodate skin grafts or flaps. Offset plates can be used to increase stability or reduce the influence of contractures on gait mechanics. Suspension systems should consider skin integrity and potential for shear damage. For example, body-powered upper extremity suspension harnesses may cause significant tissue damage over areas of graft, and a myoelectric fitting (although not typically first line) may be required. In general, suction suspension is preferred to reduce the influence of shear on compromised tissue.⁵⁴ The remainder of the prosthetic components should be determined by the survivor's functional status, based on Medicare Functional Classification K.²⁹

Hypertrophic Scarring

Cutaneous scarring after burn injury is ubiquitous and in up to 70% of survivors, resulting in visually and functionally undesirable hypertrophic scarring.²⁰ Hypertrophic scarring is a pathologic closure of burn injury, with architecture and composition distinct from normal skin due to cellular and molecular abnormalities. Hypertrophic scarring (Fig. 27.6) is characterized by elevation compared to surrounding (uninjured) skin, coloration changes that can be hyper- or hypopigmented and/or erythematous, stiff with reduced elasticity, contracted, painful, and pruritic.¹⁵ In addition, hypertrophic scarring, especially elevated hyperpigmented scar, has been associated with increased incidences of pain.¹⁶² Many consequences of hypertrophic scarring including skin fragility, dryness, open wounds, pain, itch, skin sensitivity or decreased sensation, and loss of function (contractures, tightness, decreased ROM), but can improve some persisted 2 years after injury.⁶⁵

Factors associated with increased likelihood of hypertrophic scarring include younger age; burns due to fire, flame, or flash; and markers of severity of burn injury, including percentage of TBSA



• Fig. 27.6 Hypertrophic scarring.

burned and grafted, presence of inhalation injury and length of mechanical ventilation, hospital length of stay, number of surgical interventions needed, and need for IPR.^{56,65} Burn wounds that take longer than 2 to 3 weeks to close are also predisposed to hypertrophic scarring.¹³⁶ Hypertrophic scarring is a result of imbalanced pathways required for normal healing. The process of wound healing includes hemostasis, inflammation, proliferation (including angiogenesis and extracellular matrix deposition), and remodeling.¹⁰⁹ Multiple cascades and cell signaling pathways are involved but altered. Fibroblasts deposit collagen in the proliferation phase; however, overexpression of type 1 and decreased expression of type 3 are noted in burn wounds.¹⁷³ Decreased appropriate degradation of collagen is also noted during remodeling.¹⁷³ Fibroblasts differentiate into myofibroblasts, which are responsible for wound contraction and remodeling. Myofibroblasts then also contribute multiple inflammatory cytokines to the healing milieu, perpetuating the cycle. In hypertrophic scarring, both fibroblasts and myofibroblasts are resistant to typical apoptosis during remodeling that follows appropriate wound closure.³²

Treatment options for hypertrophic burn scars may reduce but not eliminate the negative effects on cosmesis and function. Burn compression garments at 20 to 30 mm Hg compression provided within 60 days of injury, and worn consistently for 6 months, can reduce pigmentation, thickness, pain, and itch.⁹⁹ Pressure garments are thought to exert their effect by restricting blood, oxygen, and cytokine migration to the scar.¹⁶¹ The relative hypoxia leads to cellular death of fibroblasts, reduction in extracellular matrix production, and increased expression of the collagenase stimulator, prostaglandin E2, therefore degrading excess collagen.¹⁶¹ There is a new debate on the effectiveness of pressure garments in effecting long-term outcome. One must note that garments do not work over body parts that move or are soft. They are also quite expensive, and compliance rates may vary. Silicone sheeting within pressure garments may be helpful to reduce insensate water loss as dehydration.^{107,111}

Topical steroids contribute to vasoconstriction and reduction of the ongoing inflammatory cascade that contributes to hypertrophic scarring. Intralesional injection of triamcinolone historically is a mainstay of treatment, although it has variable effect and a component of undesirable effects of skin atrophy and telangiectasia. More recent studies, including an injectable combination of triamcinolone and 5-fluorouracil, showed improved scar height, pliability, and pigmentation.¹⁹²

Botulinum injections into hypertrophic scarring caused by burn injury have been promising in small studies,¹⁷⁰ with reduction in vascularity and height from unaffected skin and increased pliability. There was also benefit in symptoms of itch and pain at the scar. Multiple meta-analyses on safety and efficacy of botulinum toxin have also been favorable in reducing the appearance and characteristics of hypertrophic scarring.^{139,181} There is also described augmented benefit of combining botulinum toxin with steroid injection.^{16,101} Histopathologically, botulinum toxin is postulated to have its effect by reducing muscular tension on the wound bed, reducing fibroblast differentiation into myoblasts, and influencing downstream pathways involved in collagen production.^{10,90}

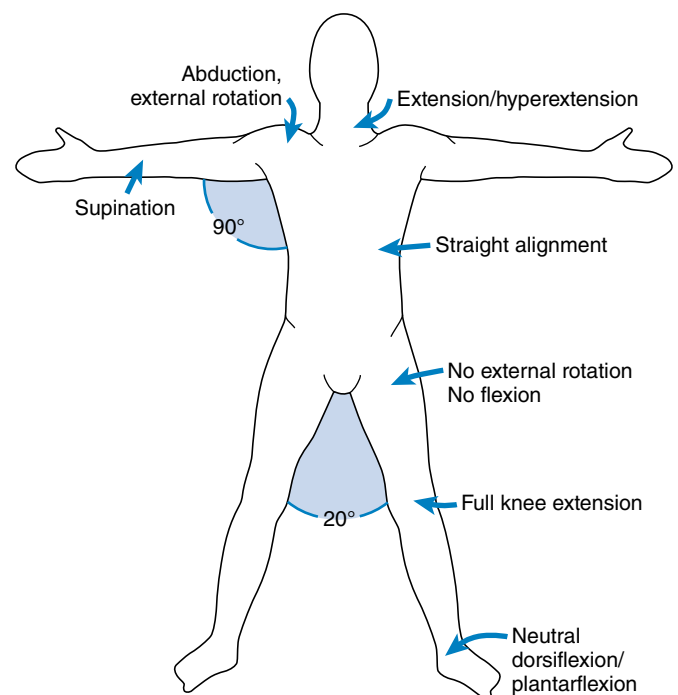
Treatment of healed burn scars with fractionated carbon dioxide (fCO₂) lasers is safe and effective in reducing visible characteristics of scar and treating associated symptoms of pain, itch, and perceived tightness.¹⁰⁶ Subsequently, measures of burn-related quality of life increased.⁸² fCO₂ lasers ablate water within abnormal collagen in scar tissue, forming microscopic holes perpendicular to the scar, which encourages remodeling.⁷⁶ fCO₂ laser

treatments are typically done on an outpatient basis.^{36,82} These treatments also decrease the number of reconstructive procedures required and are quickly becoming the treatment of choice for burn-related hypertrophic scarring.^{36,82} If the desired result with multimodal scar treatment is not achieved, surgical intervention with incisional scar-release techniques or tissue or muscle grafts can be employed.¹⁶⁶

Contracture and Prevention

Burn scar contracture is defined as a condition in which skin is replaced with pathologic scar tissue of insufficient extensibility and length resulting in loss of motion or tissue alignment of an associated joint or anatomic structure.¹²³ Contractures can alter ROM, reduce wound healing, and impair functional status, including decrement in mobility and ADL, particularly those requiring fine motor movements.⁶⁶ In order of declining prevalence, contracture most frequently develops over shoulders, elbows, wrists, ankles, and knees.⁶⁶ Maintaining a functional position while patients are in bed or immobilized both in acute care and IPR is frequently cited (Fig. 27.7). In the lower extremities, this includes keeping the hips at neutral, knees in extension, and ankles at 90 degrees of dorsiflexion. The trunk should be straight without lateral bending or flexion. Upper limb positioning should keep the shoulders abducted, elbows in approximately 30 degrees of flexion, wrists in 20 to 40 degrees of extension, with metacarpophalangeal joints in 40 to 60 degrees of flexion. The thumb should be in radial abduction. Proximal and distal interphalangeal joints should be fully extended.³⁰

Stretching is utilized to combat unopposed wound contraction and tissue shortening as burned and grafted skin heals. Gentle, slow, prolonged stretching is preferred over high-velocity stretching to reduce injury to fragile tissue.³⁰ Stretching should begin during the postoperative phase to ensure adequate tissue length to allow for full joint ROM. It should also progress to maximum tolerated for improved outcome. For example, shoulder positioning in



• Fig. 27.7 Suggested positioning guidelines to prevent contracture.

abduction greater than 110 degrees postoperatively has a better functional outcome compared to shoulder positioned at 90 degrees or less of abduction.³⁰ Stretching can be maintained by splinting. Splinting is more likely to be used for survivors with more severe injuries, as measured by longer length of stay, more hours of rehabilitation per day, more edema therapy, larger TBSA, surgical release, inhalation injury, ventilator days, and days of sedation or bedrest. Wrist, elbow, and ankle are the joints most likely to be splinted, and knees were least likely.⁹¹ However, splinting is not without controversy in the literature, given the fact that tension on a graft can induce fibroblast production of collagen and differentiation into myofibroblasts,¹⁵⁸ promoting hypertrophic scar. There is also not a consistent recommendation for type of splint or wear time, when to initiate splinting, and the degree of splinting.

The only practice guideline regarding splinting in the existing literature is that orthotic use should be considered as a treatment choice for improving ROM or reducing contracture in adults who have sustained a burn injury.¹²³ Despite advancements in multiple aspects of burn and rehabilitation care, some patients are discharged with ROM loss. The incidence of joint contractures remains between 38% and 54%,^{66,116} which is similar to nonburned patients with prolonged ICU stays.³⁵ Up to one-third of patients with a major burn leave the hospital with a contracture.⁶⁷ Skin grafting over a joint, higher TBSA necessitating grafting, male sex, concomitant medical complications, and neuropathy were more likely to develop contracture.^{66,156} However, as the time from burn injury increases, the incidence of scar contracture decreases (due to therapy, surgical correction, maturation, etc.).^{116,158}

Consideration of Inpatient Rehabilitation

Annually, 9000 burn survivors are admitted to IPR hospitals to improve their functional outcomes after burn injury.¹⁶⁹ Burn injuries are compliant with the 60% rule, designated as one of the 13 diagnoses requiring intensive IPR per Centers for Medicare and Medicaid Services.²⁷ Burn survivors in IPR demonstrate outcomes similar to those admitted due to other etiologies, including stroke, acquired brain injury, spinal cord dysfunction, orthopedic conditions, and other neurologic conditions.¹⁶⁹ The number of burn survivors discharged to IPR can vary from 8% to 26% of patients admitted to the hospital for burn injury, depending on the state where they are located and the percentage of body surface involved.⁶⁹ It is noted that IPR for these patients helps to reduce acute center length of stay and improves functional outcomes.¹⁶⁹ IPR is shown to confer a benefit to older survivors (>60 years old) compared to discharge to a skilled nursing facility (SNF)¹²⁰ and practice models should include IPR as a discharge option for older adults, considering their longer length of stay and medical comorbidities.¹²⁹ Discharge to IPR instead of SNF allows for additional time for burn healing with onsite wound care, intensive therapies to improve functional status, and thus potentially less burden of care for caregivers.¹²⁰ Depending on the functional limitations of the individual patient, a typical IPR length of stay is 2 to 4 weeks, followed by discharge to home.

Burn survivors should meet standard InterQual criteria to be considered for IPR, including ability to tolerate and participate in greater than 3 hours of therapy daily, ongoing medical issues to be managed by a physiatrist with specialized rehabilitation training, and care needs that cannot be coordinated in a subacute nursing facility environment.²⁸ Characteristics of survivors requiring comprehensive IPR services include those with more complex injuries

(including large burns [$>20\%$ TBSA], especially involving the hands and feet), ROM restriction, inhalation injury, dysphagia, and prolonged acute course (including mechanical ventilation).¹⁶⁹ Higher age, hospitalization at a burn center as opposed to an outlying hospital, and possession of health insurance were also identified as factors increasing the likelihood of IPR stay after acute hospitalization for burn injury.⁶⁹

All acute surgical interventions should be completed prior to admission to IPR to minimize interruption of the comprehensive therapy program. Burn survivors that transferred back to acute care and were subsequently readmitted to IPR had a lower functional level on admission and discharge. They were also more likely to return to acute care at final discharge, instead of to the community.⁴⁶

Therapy and Rehabilitation

IPR therapy is provided by PTs and OTs who preferably have been certified by the ABA. Burn survivors may have experienced comorbid conditions requiring immobility, such as prolonged ventilation due to inhalation or other respiratory conditions, aggressive sedation for pain management, additional traumatic injuries, or mobility and ROM restrictions after recent grafting. Physiatrists and therapists should be aware of any ROM and weight-bearing restrictions.

Early therapy focuses on elongation, stretching, maintaining ROM, contracture prevention, edema reduction, evaluation for bracing, and more dynamic activities such as transferring to a chair, maintaining sitting position, and supported standing. These activities, while not conferring a mortality benefit, are associated with reduced incidence of pneumonia, deep vein thrombosis, reduced number of hospital and ICU days, and improved peripheral and respiratory muscle strength.⁸³ In addition, early mobility lays the foundation for movement and exercise in IPR.

As the survivor progresses, treatment focus changes to active exercise, mobility, and endurance training, as well as ADL and independent ADL, independence with ROM, skin care, and caregiver training (Fig. 27.8).

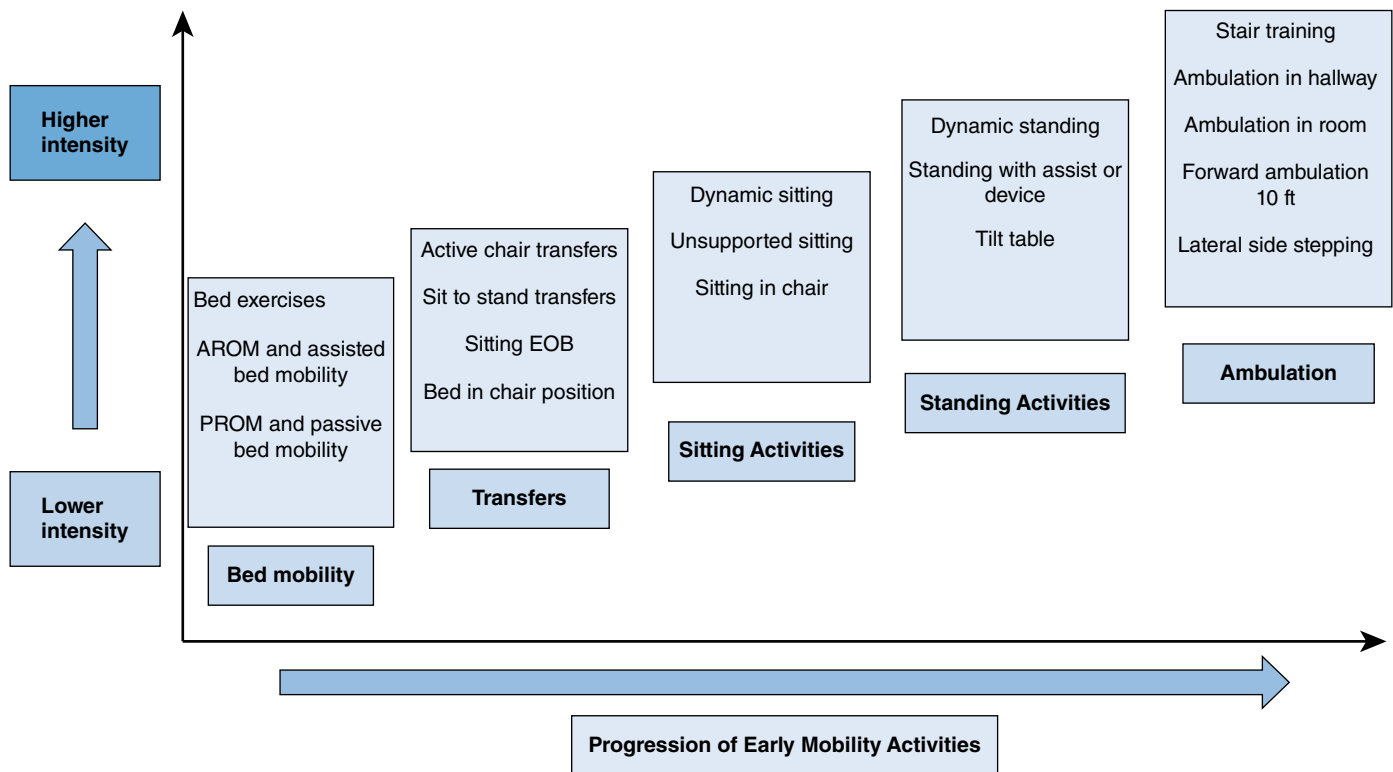
Exercise plays a vital role in the rehabilitation of patients with severe burns. Survivors experience several years of reduced exercise capacity and reduced pulmonary function following burn injury.¹³² To counteract these changes, exercise programs should be individualized for each survivor and their functional needs. The physiatrist reviews the exercise plan and progress toward goals to ensure optimal outcome. Exercise training in those surviving severe burns demonstrated improvement in lean muscle mass, strength, reduced requirement for contracture release, and improvement in physical fitness and maximal oxygen uptake.⁴⁵

Exercise programs should begin while hospitalized and continue after discharge. The ABA published guidelines for exercise prescription following burn injury, which include a resistance and/or aerobic exercise program under supervision for a minimum of 6 to 12 weeks,¹¹² progressing to independent programs.

Aerobic Training

The stress response to burns includes hypermetabolic catabolism, circulating catecholamines, cortisol, and proinflammatory factors, and results in muscle wasting and cardiac dysfunction.¹¹⁸ Survivors typically have elevated heart rate and cardiac output, as well as higher risk of cardiovascular diagnoses such as hypertension, coronary artery disease, and heart failure up to

Early Mobility Stages of Progression



• **Fig. 27.8** Early mobility stages of progression. *AROM*, Assisted range of motion; *EOB*, edge of bed; *PROM*, passive range of motion.

TABLE 27.1 Aerobic Exercise Training Guidelines

Frequency	Three to five sessions per week	
Intensity	Moderate intensity: peak 75–80% of maximum predicted heart rate (220 – age)	High intensity: peak 95% of maximum predicted heart rate
Time	Continuous moderate intensity: 20 to 40 min	High-intensity intervals: 1 to 2 min with 3 to 4 min of recovery
Volume	Combined 150 min/week	
Progression	Based on individual tolerance	
Major consideration	SAFETY with • Slow progression • Proper technique • Temperature regulation and heat dissipation • Awareness of physical limitations	

Suggested progression of aerobic exercise training after burn injury, with emphasis on individual abilities, limitations, and tolerance.

8 years after the initial injury.⁸¹ Many studies confirm reduced cardiorespiratory fitness in survivors compared to age-matched controls.^{45,57,112}

Recommendations for aerobic exercise in adult and pediatric burn survivors include a 12-week rehabilitation exercise training protocol, beginning at discharge from the hospital.¹¹⁸ The protocol can be initiated as outpatient therapy or as part of a comprehensive IPR program. Each aerobic session should be tailored to the individual's current functional status and capabilities, while ensuring maximum safety. Typical aerobic protocol includes a heart rate of 70% to 85% of predicted max (220 – age) initially

for only 5 to 10 minutes, with progression to 20 to 40 minutes, three to five times per week. As patients improve, small bursts of more intense exercise intervals (95% of predicted maximum heart rate) for a few minutes followed by a prolonged rest may be incorporated. Aerobic activity to achieve heart rate goals should be dynamic and incorporate large muscles of the legs, back, buttocks, core, and arms. Understanding that the level of deconditioning in survivors may be profound, easy walking may be enough activity initially to achieve the heart rate target. Treadmill, stationary bike, arm ergometer, and other exercise machines may be incorporated as aerobic capacity increases (Table 27.1).

TABLE 27.2 Resistance Exercise Recommended Progression

Weeks	% of Maximum Weight	Sets / Repetitions
1–3	50–60%	3 / 10 to 15
4–6	70–75%	3 / 8 to 10
6–11	75–80%	3 / 8 to 12

Suggested progression of resistance exercises during a 12-week program. After 12 weeks, weight or repetitions can be increased by 10% each week.

Resistance Training

The safety of adding resistance training in burn survivors within the acute care setting has been established.⁶² The addition of resistance training or sham resistance training in two matched and compared groups found no differences in adverse outcomes or increased length of stay due to complications. There was improvement in the strength of the upper extremities in those who received resistance training compared to the standard PT group.⁶² Since safety and efficacy have been established in the acute care setting, presumably when survivors are more ill, the safety and efficacy of exercise in IPR and as an outpatient is implied.

Resistance training programs are used to strengthen individual muscles using machines, bands, free weights, or body weights.^{62,118} Therapists should provide initial instruction on form, technique, and safety, considering each individual’s pain and current functional level.⁶² Typical resistance exercise programs in burn survivors target large muscle groups crossing multiple joints (such as legs, chest, back) or isolate specific movements (such as biceps curls or triceps press). Strengthening of the core should also be incorporated. Gradual progression is recommended (Table 27.2) with further increases after 12 weeks, by increasing the weight or repetitions by 5% to 10% each week.¹¹⁸ This program can be initiated in the IPR setting, with instructions for continuation during outpatient PT or as a self-directed program.

Psychological Considerations of Survivorship

Screening and treatment of burn survivors’ psychological state is paramount. Burn survivors endorse preinjury mental health concerns at rates higher than the general population,^{53,126} particularly with depression and substance use disorder,¹⁰² which can be compounded by postinjury fear, anxiety, pain, delirium, and grief.¹⁷⁵ Additionally, even years after initial injury, survivors report decreased quality of life and poorer physical and mental health^{3,17} compared to nonburned individuals, making screening and treatment for psychological concerns essential parts of recovery.

Recommendations from the ABA Psychological Outcomes Consensus Committee in 2013 advocate screening for psychological outcomes as a quality metric to focus on data collection and elevation of psychosocial care for burn survivors.⁶⁰ In addition, recommendations include screening for depression and acute stress disorder (ASD) symptoms within 48 hours of the survivors’ ability to participate in the interview, and at least once prior to discharge from the hospital. When returning for outpatient care, all survivors should have repeat screening (for depression, ASD, and posttraumatic stress disorder [PTSD]) at their first clinic visit and subsequent visits as needed. All ABA-certified

burn centers are required to outline their process for referral when survivors screen positive.⁶⁰

Delirium

Acutely, ICU delirium is characterized by change in baseline cognition and psychomotor status, with fluctuations in intensity and severity throughout the day, not attributable to a separate medical condition such as neurologic, infectious, withdrawal symptoms, hypoxia, and the like.⁸ Delirium is common among burn survivors, with an incidence of 47% of those admitted to the ICU.¹ This increases to 77% in those who also require ventilation in the ICU setting.⁴ Interventions to reduce harm from delirium (including mortality and length of ICU stay)¹² center on restorative day/night scheduling, increased daytime mobility, reorientation, presence of familiar family members and friends, optimization of pain management, and minimizing deliriogenic medications.¹² Judicious use of medications to treat symptoms of delirium, such as paranoia, agitation, risk of dislodging medical devices, or treatments, and harm to the survivor and staff can be considered.¹⁴³

Anxiety

Anxiety and pain are often an interlinked and perpetuating cycle, whereby anxiety in anticipation of pain affects the perception of pain,^{85,168} and the presence of pain increases anxiety. Interventions can include medicating prior to procedural interventions and frequent provision of information about expected schedules, daily activities, procedures, and outcomes.¹⁶⁸ Anxiety in burn survivors may also be a direct manifestation of the trauma itself, as well as concerns about financial stability, and survivors questioning their ability to reengage in family, work, and social roles. Anxiety tends to be highest in the acute phase but remains present in at least 20% of survivors at 6 months. Long-term symptoms directly related to the injury can persist for up to 2 years.¹²¹

Depression

Postburn injury depression is difficult to quantify in the literature due to inconsistent measurement methods and reporting.^{177,178} However, depression postburn injury can be present acutely, triggered by realization of consequences of injury, including functional limitations and altered sense of self. It can be closely correlated with survivors’ preinjury incidence of depression.¹⁰² Rates of in-hospital depression range from 53% to 15% and tend to be more severe acutely and decrease as time from injury elapses.¹³⁷ At 1 year postinjury, 13% to 23% of survivors report symptoms consistent with major depression.¹⁷⁸ A greater proportion endorse psychological distress without strictly meeting the diagnostic criteria for depression. Importantly, depression after burn injury has been found to correlate with decreased physical functioning 1 year after injury.¹⁷⁶ Treatment cornerstones include CBT and medications.

Acute Stress Disorder and Posttraumatic Stress Disorder

ASD presents as hyperarousal and reexperiencing negative events with concomitant negative changes in cognition and mood within 3 to 10 days of onset of exposure or experience of a traumatic event.⁸ Prevalence in burn survivors ranges from 10% to 23%⁷⁶ and is predicted by female sex, presence of visible injury, and

preinjury psychological status. ASD within the hospital setting is predictive of PTSD symptoms up to 2 years postinjury.⁵⁹ Symptoms of PTSD include thought intrusions (distressing memories or dreams, flashbacks), avoidance of memories, stimuli or reminders of the trauma, and negative changes in mood such as irritability, increased startle response, difficulty with sleep, and feelings of blame and isolation.⁸ These symptoms must present after exposure or experience of a trauma, last more than 1 month, and lead to distress or functional deficits. PTSD has a higher incidence in burn survivors than in the trauma population in general, with rates recorded as 15% to 45% 1 year following trauma.^{59,178} There is an association between higher levels of posttraumatic stress and chronic pain and itch.¹⁷⁸ Treatment includes trauma-informed psychological therapy (CBT, group therapy), eye movement desensitization and reprocessing, and medications.

Sleep Disturbance

Due to physical discomfort such as pain, itch, and emotional distress, it is not surprising that many survivors have disturbance in sleep. The prevalence is 40% to 73% and may persist for months to years following burn injury.⁹⁷ Sleep hygiene practice information and medication interventions can be helpful.¹⁵³

Body Image and Appearance Satisfaction

Coping with changes in appearance, body image, and the inevitable effect these have on self-esteem and sexuality remain a great challenge for burn survivors,¹⁸⁴ especially females and those who place great value on personal appearance and attractiveness.⁹⁵ Female sex, higher TBSA burn, and importance of appearance predict body dissatisfaction of survivors, which in turn affects psychosocial function at 1 year.¹⁷² Circularly, psychosocial components, including depression, self-acceptance, and comfort with social integration, are also important factors in body satisfaction after burn.⁹⁵ Therefore routine psychological screening, including body image and distress, should be utilized both in the hospital and at follow-up visits and appropriate referrals made.¹⁷²

Sexuality and Sexual Functioning

Sexual functioning is intricately tied to body appearance and satisfaction and should be inquired about by the comprehensive burn team early in care utilizing open-ended questions.^{122,175} Unfortunately, only a small proportion (14%) of practitioners feel comfortable and qualified to initiate this discussion.¹⁴⁵ However, all practitioners working with burn survivors and their partners should be aware that concerns about sexuality and sexual functioning are common and important to them and can have long-term impact on quality of life.³⁷ The changes brought about by burn injury affect sexual function in survivors. For example, scars leading to decreased sensitivity or hypersensitivity of skin, pain, changes to ROM, or damage to genitalia can make sexual function challenging.³⁹ Fear or rejection or body image concerns can also affect sexual confidence. Side effects from medications, such as beta blockers or selective serotonin reuptake inhibitors, can decrease libido.³⁹ Circulating catecholamines and hypermetabolic hormones may interfere with sex hormone release.¹²² Practitioners working with burn survivors should be aware of these effects on sexual functioning and be willing to initiate the conversation and provide therapeutic guidance, whether from OTs to optimize comfort and safety with positioning, physicians making medication

adjustments, or psychologists to address body acceptance and communication.¹⁷⁵ Referrals to online resources such as the Phoenix Society for Burn Survivors or in-person or virtual specialized sex therapists or psychiatrists should also be within the realm of burn practitioners.¹³¹

Social and Community Reintegration

Burn survivors cite social anxiety and strain of interacting with those who do not understand the long-term sequelae of burn, while they themselves are coping with physical and psychological changes, to be the most difficult part of community reintegration.¹⁹ Reintegration is particularly challenging for those with facial involvement¹⁴⁶ and those with poor physical function, who even 5 years after injury report feelings of depression, social isolation, and loss of peer group.¹¹⁹ Survivors also have difficulty meeting new friends, making friends, and forging intimate relationships.¹⁴⁹

Programs to assist with acceptance of appearance, self-esteem, and social skills following burn injury are also excellent resources to assist with community integration after burn. The Phoenix Society for Burn Survivors has developed multiple programs to facilitate social and community reintegration, including Survivors Offering Assistance with Recovery (SOAR), where screened and trained survivors provide peer mentorship, knowledge, empathy, and support to those more recently injured. SOAR mentors provide support throughout recovery in one-on-one visits, group meetings online or in person, live chats, and social media postings.¹³¹ Other programs developed by the Phoenix Society include Beyond Surviving: Tools for Thriving After Burn Injury, which promotes life strategies to be comfortable, confident, and competent in any social, work, or school situation. Changing Faces is another program designed to facilitate social skills, self-esteem, and community integration using the 3-2-1-GO model, which encourages survivors to practice how they will respond to others' reactions to their appearance.¹²⁴ In the 3-2-1-Go model, survivors compose three things they will do if someone stares at them, two answers if someone asks what happened, and one thing to think if someone turns away. These programs help survivors cope with self-consciousness that may occur with public reactions to their changed appearance and promote self-efficacy.

Return to Work

For survivors of employment age, return to work is one signifier of recovery. Analysis of the Burn Model Systems National Database demonstrated that those employed preburn are most likely to return to work in the 12 months postburn.^{25,189} Survivors who participated in IPR programs had a higher likelihood of returning to employment 1 year postinjury compared to those who received rehabilitation at a SNF, long-term care hospital, or other extended care facility, despite a more severe injury.⁵⁰ White, non-Hispanic survivors were more likely to be employed postburn than other ethnic groups.^{25,189} Multiple published reviews have identified factors associated with unemployment postburn (in those who were employed at the time of injury), including older age, female sex, longer inpatient hospitalization, elevated levels of pain, and government (vs. private) insurance.^{25,103,140} Those who were injured at work were twice as likely to be unemployed at 1 year as those burned outside of work.¹⁵⁴ Other studies have determined psychological factors, such as premorbid anxiety, depression, and alcoholism, and amputation status are also important predictors of return to work.^{26,52} The type of employment and the physical

demands thereof affected the time to return to work.²¹ Those with fewer physical requirements for their job were able to return more quickly. Additionally, survivors who developed contractures at the time of hospital discharge were also less likely to be employed postburn than those who did not have contracture formation.¹³⁰ There is conflicting conclusion in the literature as to whether return to work status was influenced by the size of the burn based on TBSA, the involvement (and to what degree) of hands,^{21,140} and whether workers' compensation was utilized.^{25,189} These inconsistencies suggest that the psychological factors associated with return to work play a much larger role than physical factors. The most common barriers to returning to work include pain, neurologic symptoms, and impaired mobility, which are most typically seen in more severe injury.¹⁵⁴ Only 37% of those who were able to return to work could engage in the same job without accommodations,²³ further underscoring the importance of physiatrists in the return-to-work process.

Summary

Burn rehabilitation is a lifelong process for survivors. Physiatrists are uniquely positioned to work as part of an interdisciplinary team, identifying and managing complications of burn injury across the care spectrum, including inpatient, IPR, and outpatient stages of recovery, to reduce the systemic impact of the injury and improve quality of life. Burn physiatrists implement interventions for pain, itch, neuropathy, delirium, psychosocial coping and mood, and nutritional needs. Burn physiatrists also oversee and direct a rehabilitation program focusing on scar management, contracture prevention, dysphagia treatment, evaluation and treatment of cognitive impairments, and focus on aerobic and resistive exercises to optimize functional recovery. Burn physiatrists can also be instrumental in providing resources for coping with changing abilities and perceptions of self, while focusing on community reintegration and return to work.

Key References

2. Abdullahi A, Jeschke MG. Nutrition and anabolic pharmacotherapies in the care of burn patients. *Nutr Clin Pract*. 2014;29(5):621-630.
3. Abouzeid CA, Wolfe AE, Ni P, et al. Are burns a chronic condition? Examining patient reported outcomes up to 20 years after burn injury—a Burn Model System National Database investigation. *J Trauma Acute Care Surg*. 2022;92(6):1066-1074.
9. Arnoldo BD, Purdue GF, Kowalske K, et al. Electrical injuries: a 20-year review. *J Burn Care Rehabil*. 2004;25(6):479-484.
11. Bajorek AJ, Slocum C, Goldstein R, et al. Impact of cognition on burn inpatient rehabilitation outcomes. *PM&R*. 2017;9(1):1-7.
13. Bartley CN, Atwell K, Purcell L, et al. Amputation following burn injury. *J Burn Care Res*. 2019;40(4):430-436.
17. Bich CS, Kostev K, Baus A, et al. Burn injury and incidence of psychiatric disorders: a retrospective cohort study of 18,198 patients from Germany. *Burns*. 2021;47(5):1110-1117.

21. Bowden ML, Thomson PD, Prasad JK. Factors influencing return to employment after a burn injury. *Arch Phys Med Rehabil*. 1989;70(10):772-774.
24. Carmichael H, Wiktor A, Wagner A, et al. High risk of developing long-term opioid use after burn injury. *J Am Coll Surg*. 2018;227(4/1):S264.
26. Carrougier GJ, Martinez EM, McMullen KS, et al. Pruritus in adult burn survivors: postburn prevalence and risk factors associated with increased intensity. *J Burn Care Res*. 2013;34(1):94-101.
36. Clayton NA, Ward EC, Rumbach AF, et al. Influence of inhalation injury on incidence, clinical profile and recovery pattern of dysphagia following burn injury. *Dysphagia*. 2020;35(6):968-977.
47. Donthula D, Conner CR, Truong VTT, et al. Impact of opioid-minimizing pain protocols after burn injury. *J Burn Care Res*. 2021;42(6):1146-1151.
53. Fauerbach JA, Lawrence J, Haythornthwaite J, et al. Preburn psychiatric history affects posttrauma morbidity. *Psychosomatics*. 1997;38(4):374-385.
57. Ganio MS, Pearson J, Schlader ZJ, et al. Aerobic fitness is disproportionately low in adult burn survivors years after injury. *J Burn Care Res*. 2015;36(4):513-519.
62. Gittings PM, Wand BM, Hince DA, et al. The efficacy of resistance training in addition to usual care for adults with acute burn injury: a randomised controlled trial. *Burns*. 2021;47(1):84-100.
71. Gurbuz K, Demir M, Basaran A, et al. Most prominent factors contributing to burn injury-related amputations: an analysis of a referral burn center. *J Burn Care Res*. 2022;43(4):921-925.
75. Hendricks CT, Camara K, Violick Boole K, et al. Burn injuries and their impact on cognitive-communication skills in the inpatient rehabilitation setting. *J Burn Care Res*. 2017;38(1):e359-e369.
94. Kneib CJ, Sibbett SH, Carrougier GJ, et al. The effects of early neuropathic pain control with gabapentin on long-term chronic pain and itch in burn patients. *J Burn Care Res*. 2019;40(4):457-463.
102. Logsetty S, Shamlou A, Gawaziuk JP, et al. Mental health outcomes of burn: a longitudinal population-based study of adults hospitalized for burns. *Burns*. 2016;42(4):738-744.
112. Nedelec B, Parry I, Acharya H, et al. Practice guidelines for cardiovascular fitness and strengthening exercise prescription after burn injury. *J Burn Care Res*. 2016;37(6):e539-e558.
118. Palackic A, Suman OE, Porter C, et al. Rehabilitative exercise training for burn injury. *Sports Med*. 2021;51(12):2469-2482.
169. Tan WH, Goldstein R, Gerrard P, et al. Outcomes and predictors in burn rehabilitation. *J Burn Care Res*. 2012;33(1):110-117.
178. Van Loey NEE, de Jong AEE, Hofland HWC, et al. Role of burn severity and posttraumatic stress symptoms in the co-occurrence of itch and neuropathic pain after burns: a longitudinal study. *Front Med (Lausanne)*. 2022;9:997183.
180. Villarreal E, Wolf SE, Golovko G, et al. Opioid prescription and opioid disorders in burns: a large database analysis from 1990 to 2019. *Burns*. 2023;49(8):1845-1853.
189. Wrigley M, Trotman BK, Dimick A, et al. Factors relating to return to work after burn injury. *J Burn Care Rehabil*. 1995;16(4):445-450, discussion 444.

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References

- Abdelrahman I, Vieweg R, Irschik S, et al. Development of delirium: association with old age, severe burns, and intensive care. *Burns*. 2020;46(4):797-803.
- Abdullahi A, Jeschke MG. Nutrition and anabolic pharmacotherapies in the care of burn patients. *Nutr Clin Pract*. 2014;29(5):621-630.
- Abouzeid CA, Wolfe AE, Ni P, et al. Are burns a chronic condition? Examining patient reported outcomes up to 20 years after burn injury—a Burn Model System National Database investigation. *J Trauma Acute Care Surg*. 2022;92(6):1066-1074.
- Agarwal V, O'Neill PJ, Cotton BA, et al. Prevalence and risk factors for development of delirium in burn intensive care unit patients. *J Burn Care Res*. 2010;31(5):706-715.
- Ahuja RB, Gupta R, Gupta G, et al. A comparative analysis of cetirizine, gabapentin and their combination in the relief of post-burn pruritus. *Burns*. 2011;37(2):203-207.
- American Burn Association. *Advanced Burn Life Support Course Provider Manual*. Accessed July 5, 2024. <https://ameriburn.org/wp-content/uploads/2019/08/2018-abls-providermanual.pdf>.
- American Burn Association. About. Accessed July 1, 2024. ameriburn.org.
- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed. APA; 2013.
- Arnold BD, Purdue GF, Kowalske K, et al. Electrical injuries: a 20-year review. *J Burn Care Rehabil*. 2004;25(6):479-484.
- Austin E, Koo E, Jagdeo J. The cellular response of keloids and hypertrophic scars to botulinum toxin A: a comprehensive literature review. *Dermatol Surg*. 2018;44(2):149-157.
- Bajorek AJ, Slocum C, Goldstein R, et al. Impact of cognition on burn inpatient rehabilitation outcomes. *PM&R*. 2017;9(1):1-7.
- Barr J, Fraser GL, Puntillo K, et al. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med*. 2013;41(1):263-306.
- Bartley CN, Atwell K, Purcell L, et al. Amputation following burn injury. *J Burn Care Res*. 2019;40(4):430-436.
- Bermo MS, Patterson D, Sharar SR, et al. Virtual reality to relieve pain in burn patients undergoing imaging and treatment. *Top Magn Reson Imaging*. 2020;29(4):203-208.
- Bharadia SK, Burnett L, Gabriel V. Hypertrophic scar. *PM&R*. 2023;34(4):783-798.
- Bi M, Sun P, Li D, et al. Intralesional injection of botulinum toxin type A compared with intralesional injection of corticosteroid for the treatment of hypertrophic scar and keloid: a systematic review and meta-analysis. *Med Sci Monit*. 2019;25:2950-2958.
- Bich CS, Kostev K, Baus A, et al. Burn injury and incidence of psychiatric disorders: a retrospective cohort study of 18,198 patients from Germany. *Burns*. 2021;47(5):1110-1117.
- Bittner EA, Shank E, Woodson L, et al. Acute and perioperative care of the burn-injured patient. *Anesthesiology*. 2015;122(2):448-464.
- Blakeney P, Partridge J, Rumsey N. Community integration. *J Burn Care Res*. 2007;28(4):598-601.
- Bombaro KM, Engrav LH, Carrougner GJ, et al. What is the prevalence of hypertrophic scarring following burns? *Burns*. 2003;29(4):299-302.
- Bowden ML, Thomson PD, Prasad JK. Factors influencing return to employment after a burn injury. *Arch Phys Med Rehabil*. 1989;70(10):772-774.
- Breederveld RS, Tuinebreijer WE. Recombinant human growth hormone for treating burns and donor sites. *Cochrane Database Syst Rev*. 2012;12:CD008990.
- Brych SB, Engrav LH, Rivara FP, et al. Time off work and return to work rates after burns: systematic review of the literature and a large two-center series. *J Burn Care Rehabil*. 2001;22(6):401-405.
- Carmichael H, Wiktor A, Wagner A, et al. High risk of developing long-term opioid use after burn injury. *J Am Coll Surg*. 2018;227(4/1):S264.
- Carrougner GJ, Bamer AM, Mandell SP, et al. Factors affecting employment after burn injury in the United States: a Burn Model System National Database investigation. *Arch Phys Med Rehabil*. 2020;101(1):S71-S85.
- Carrougner GJ, Martinez EM, McMullen KS, et al. Pruritus in adult burn survivors: postburn prevalence and risk factors associated with increased intensity. *J Burn Care Res*. 2013;34(1):94-101.
- Centers for Medicare and Medicaid Services. *Centers for Medicare and Medicaid Services Presumptive Compliance-3 (ICD-10-CM)*. Accessed July 1, 2024. [https://www.cms.gov/medicare/payment/prospective-payment-systems/inpatient-rehabilitation/data-files#:~:text=Presumptive%20Compliance%2D3%20\(ICD%2D~start%20of%20the%20fiscal%20year](https://www.cms.gov/medicare/payment/prospective-payment-systems/inpatient-rehabilitation/data-files#:~:text=Presumptive%20Compliance%2D3%20(ICD%2D~start%20of%20the%20fiscal%20year).
- Centers for Medicare & Medicaid Services. *Fact Sheet #1: Inpatient Rehabilitation Facility Classification Requirements*. <https://www.cms.gov/Medicare/Medicare-Fee-for-Service/Payment/InpatientRehabFacPPS/Downloads/fs1classreq.pdf>.
- CMS. *Centers for Medicare and Medicaid Services Health and Technology Assessment. Lower Limb Prosthetic Work Group Consensus Document*. https://www.cms.gov/medicare/coverage/determinationprocess/downloads/llp_consensus_document.pdf.
- Chapman TT. Burn scar and contracture management. *J Trauma*. 2007;62(6):S8.
- Cheah AKW, Kangkorn T, Tan EH, et al. The validation study on a three-dimensional burn estimation smart-phone application: accuracy, free and fast? *Burns Trauma*. 2018;6:7.
- Chiang RS, Borovikova AA, King K, et al. Current concepts related to hypertrophic scarring in burn injuries. *Wound Repair Regen*. 2016;24(3):466-477.
- Chou R, Gordon DB, de Leon-Casasola OA, et al. Management of postoperative pain: a clinical practice guideline from the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council. *J Pain*. 2016;17(2):131-157.
- Chung BY, Kim HB, Jung MJ, et al. Post-burn pruritus. *Int J Mol Sci*. 2020;21(11):3880.
- Clavet H, Hebert PC, Fergusson D, et al. Joint contracture following prolonged stay in the intensive care unit. *CMAJ*. 2008;178(6):691-697.
- Clayton NA, Ward EC, Rumbach AF, et al. Influence of inhalation injury on incidence, clinical profile and recovery pattern of dysphagia following burn injury. *Dysphagia*. 2020;35(6):968-977.
- Connell KM, Phillips M, Coates R, et al. Sexuality, body image and relationships following burns: analysis of BSHS-B outcome measures. *Burns*. 2014;40(7):1329-1337.
- Crawford CM, Varghese G, Mani MM, et al. Heterotopic ossification: are range of motion exercises contraindicated? *J Burn Care Rehabil*. 1986;7(4):323-327.
- de Rios MD, Novac A, Achauer BH. Sexual dysfunction and the patient with burns. *J Burn Care Rehabil*. 1997;18(1 Pt 1):37-42.
- Deitch EA, Wheelahan TM, Rose MP, et al. Hypertrophic burn scars: analysis of variables. *J Trauma*. 1983;23(10):895-898.
- Demling RH. Comparison of the anabolic effects and complications of human growth hormone and the testosterone analog, oxandrolone, after severe burn injury. *Burns*. 1999;25(3):215-221.
- Demling RH, Orgill DP. The anticatabolic and wound healing effects of the testosterone analog oxandrolone after severe burn injury. *J Crit Care*. 2000;15(1):12-17.
- Deng H, Chen J, Li F, et al. Effects of mobility training on severe burn patients in the BICU: a retrospective cohort study. *Burns*. 2016;42(7):1404-1412.
- Deveci M, Bozkurt M, Sengezer M. Clonus: an unusual delayed neurological complication in electrical burn injury. *Burns*. 2001;27(6):647-651.
- Disseldorp LM, Nieuwenhuis MK, Van Baar ME, et al. Physical fitness in people after burn injury: a systematic review. *Arch Phys Med Rehabil*. 2011;92(9):1501-1510.
- DiVita MA, Mix JM, Goldstein R, et al. Rehabilitation outcomes among burn injury patients with a second admission to an inpatient rehabilitation facility. *PM&R*. 2014;6(11):999-1007.
- Donthula D, Conner CR, Truong VTT, et al. Impact of opioid-minimizing pain protocols after burn injury. *J Burn Care Res*. 2021;42(6):1146-1151.

48. Dougherty W, Waxman K. The complexities of managing severe burns with associated trauma. *Surg Clin North Am.* 1996;76(4):923-958.
49. Emery MA, Eitan S. Drug-specific differences in the ability of opioids to manage burn pain. *Burns.* 2020;46(3):503-513.
50. Espinoza LF, Simko LC, Goldstein R, et al. Postacute care setting is associated with employment after burn injury. *Arch Phys Med Rehabil.* 2019;100(11):2015-2021.
51. Esselman PC. Burn rehabilitation: an overview. *Arch Phys Med Rehabil.* 2007;88(12/2):S3-S6.
52. Fauerbach JA, Heinberg LJ, Lawrence JW, et al. Coping with body image changes following a disfiguring burn injury. *Health Psychol.* 2002;21(2):115-121.
53. Fauerbach JA, Lawrence J, Haythornthwaite J, et al. Preburn psychiatric history affects posttrauma morbidity. *Psychosomatics.* 1997;38(4):374-385.
54. Ferguson JR, Blanck R. Prosthetic management of the burn amputation. *PM&R.* 2011;22(2):277-299, vi.
55. Gabriel V, Kowalske KJ, Holavanahalli RK. Assessment of recovery from burn-related neuropathy by electrodiagnostic testing. *J Burn Care Res.* 2009;30(4):668-674.
56. Gangemi EN, Gregori D, Berchialli P, et al. Epidemiology and risk factors for pathologic scarring after burn wounds. *Arch Facial Plast Surg.* 2008;10(2):93-102.
57. Gano MS, Pearson J, Schlader ZJ, et al. Aerobic fitness is disproportionately low in adult burn survivors years after injury. *J Burn Care Res.* 2015;36(4):513-519.
58. Gawaziuk JP, Peters B, Logsetty S. Early ambulation after-grafting of lower extremity burns. *Burns.* 2018;44(1):183-187.
59. Giannoni-Pastor A, Eiroa-Orosa FJ, Fidel Kinori SG, et al. Prevalence and predictors of posttraumatic stress symptomatology among burn survivors: a systematic review and meta-analysis. *J Burn Care Res.* 2016;37(1):e79-e89.
60. Gibran NS, Wiechman S, Meyer W, et al. Summary of the 2012 ABA Burn Quality Consensus conference. *J Burn Care Res.* 2013;34(4):361-385.
61. Gillman A, Winkler R, Taylor NF. Implementing the free water protocol does not result in aspiration pneumonia in carefully selected patients with dysphagia: a systematic review. *Dysphagia.* 2017;32(3):345-361.
62. Gittings PM, Wand BM, Hince DA, et al. The efficacy of resistance training in addition to usual care for adults with acute burn injury: a randomised controlled trial. *Burns.* 2021;47(1):84-100.
63. Godleski M, Yelvington M, Jean S. Burn injury complications impacting rehabilitation. *PM&R.* 2023;34(4):799-809.
64. Goutos I, Eldardiri M, Khan AA, et al. Comparative evaluation of antipruritic protocols in acute burns. The emerging value of gabapentin in the treatment of burns pruritus. *J Burn Care Res.* 2010;31(1):57-63.
65. Goverman J, He W, Martello G, et al. The presence of scarring and associated morbidity in the Burn Model System National Database. *Ann Plast Surg.* 2019;82(3/2):S162-S168.
66. Goverman J, Mathews K, Goldstein R, et al. Adult contractures in burn injury: a Burn Model System National Database study. *J Burn Care Res.* 2017;38(1):e328-e336.
67. Governman J, Mathews K, Goldstein R, et al. Adult contractures in burn injury: a burn model systems national database study. *J Burn Care Res.* 2017;38(1):e328-e336.
68. Gray P, Williams B, Cramond T. Successful use of gabapentin in acute pain management following burn injury: a case series. *Pain Med.* 2008;9(3):371-376.
69. Greene NH, Pham TN, Esselman PC, et al. Variation in inpatient rehabilitation utilization after hospitalization for burn injury in the United States. *J Burn Care Res.* 2015;36(6):613-618.
70. Greenhalgh DG. Management of burns. *N Engl J Med.* 2019;380(24):2349-2359.
71. Gurbuz K, Demir M, Basaran A, et al. Most prominent factors contributing to burn injury-related amputations: an analysis of a referral burn center. *J Burn Care Res.* 2022;43(4):921-925.
72. Handschin AE, Vetter S, Jung FJ, et al. A case-matched controlled study on high-voltage electrical injuries vs thermal burns. *J Burn Care Res.* 2009;30(3):400-407.
73. Hart DW, Wolf SE, Mlcak R, et al. Persistence of muscle catabolism after severe burn. *Surgery.* 2000;128(2):312-319.
74. Haruta A, Mandell SP. Assessment and management of acute burn injuries. *PM&R.* 2023;34(4):701-716.
75. Hendricks CT, Camara K, Violick Boole K, et al. Burn injuries and their impact on cognitive-communication skills in the inpatient rehabilitation setting. *J Burn Care Res.* 2017;38(1):e359-e369.
76. Herndon DN. *Total Burn Care.* 5th ed. Elsevier; 2018:xix.
77. Hill DM, DeBoer E. State and future science of opioids and potential of biased-ligand technology in the management of acute pain after burn injury. *J Burn Care Res.* 2023;44(3):524-534.
78. Hill DM, Ly A, Desai JP, et al. Efficacy of a novel LAM femoral cutaneous block technique for acute donor site pain. *J Burn Care Res.* 2023;44(1):16-21.
79. Holmes IV JH, Molnar JA, Carter JE, et al. A comparative study of the ReCell device and autologous spit-thickness meshed skin graft in the treatment of acute burn injuries. *J Burn Care Res.* 2018;39(5):694-702.
80. Holtman Jr JR, Jellish WS. Opioid-induced hyperalgesia and burn pain. *J Burn Care Res.* 2012;33(6):692-701.
81. Hung TY, Lee YK, Huang MY, et al. Increased risk of ischemic stroke in patients with burn injury: a nationwide cohort study in Taiwan. *Scand J Trauma Resusc Emerg Med.* 2016;24:44.
82. Issler-Fisher AC, Fisher OM, Haertsch PA, et al. Effectiveness and safety of ablative fractional CO(2) laser for the treatment of burn scars: a case-control study. *Burns.* 2021;47(4):785-795.
83. Jacobson K, Fletchall S, Dodd H, et al. Current concepts burn rehabilitation, part I: care during hospitalization. *Clin Plast Surg.* 2017;44(4):703-712.
84. James DL, Jowza M. Principles of burn pain management. *Clin Plast Surg.* 2017;44(4):737-747.
85. Jones LM, Uribe AA, Coffey R, et al. Pregabalin in the reduction of pain and opioid consumption after burn injuries: a preliminary, randomized, double-blind, placebo-controlled study. *Medicine (Baltimore).* 2019;98(18):e15343.
86. Kagan RJ, Peck MD, Ahrenholz DH, et al. Surgical management of the burn wound and use of skin substitutes: an expert panel white paper. *J Burn Care Res.* 2013;34(2):e60-e79.
87. Kamolz LP, Kitzinger HB, Karle B, et al. The treatment of hand burns. *Burns.* 2009;35(3):327-337.
88. Kaul I, Amin A, Rosenberg M, et al. Use of gabapentin and pregabalin for pruritus and neuropathic pain associated with major burn injury: a retrospective chart review. *Burns.* 2018;44(2):414-422.
89. Kennedy PJ, Young WM, Deva AK, et al. Burns and amputations: a 24-year experience. *J Burn Care Res.* 2006;27(2):183-188.
90. Khateri BHM, Hussein HA, Abd-El-Raheem TA, et al. Assessment of intralesional injection of botulinum toxin type A in hypertrophic scars and keloids: clinical and pathological study. *Dermatol Ther.* 2022;35(10):e15748.
91. Khor D, Liao J, Fleishhacker Z, et al. Update on the practice of splinting during acute burn admission from the ACT study. *J Burn Care Res.* 2022;43(3):640-645.
92. Kim DE, Pruskowski KA, Ainsworth CR, et al. A review of adjunctive therapies for burn injury pain during the opioid crisis. *J Burn Care Res.* 2019;40(6):983-995.
93. Klein MB, Lezotte DL, Fauerbach JA, et al. The National Institute on Disability and Rehabilitation Research burn model system database: a tool for the multicenter study of the outcome of burn injury. *J Burn Care Res.* 2007;28(1):84-96.
94. Kneib CJ, Sibbett SH, Carrougher GJ, et al. The effects of early neuropathic pain control with gabapentin on long-term chronic pain and itch in burn patients. *J Burn Care Res.* 2019;40(4):457-463.
95. Lawrence JW, Fauerbach JA, Thombs BD. A test of moderating role of importance of appearance in the relationship between perceived scar severity and body-esteem among adult burn survivors. *Body Image.* 2006;3:101-111.

96. Lee HG, Grossman SK, Valdes-Rodriguez R, et al. Topical ketamine-amitriptyline-lidocaine for chronic pruritus: a retrospective study assessing efficacy and tolerability. *J Am Acad Dermatol*. 2017;76(4):760-761.
97. Lerman SF, Owens MA, Liu T, et al. Sleep after burn injuries: a systematic review and meta-analysis. *Sleep Med Rev*. 2022;65:101662.
98. Levi B, Jayakumar P, Giladi A, et al. Risk factors for the development of heterotopic ossification in seriously burned adults: a National Institute on Disability, Independent Living and Rehabilitation Research burn model system database analysis. *J Trauma Acute Care Surg*. 2015;79(5):870-876.
99. Li P, Li-Tsang CWP, Deng X, et al. The recovery of post-burn hypertrophic scar in a monitored pressure therapy intervention programme and the timing of intervention. *Burns*. 2018;44(6):1451-1467.
100. Lin S-H, Chou C-L, Chiou H-J. Ultrasonography in early diagnosis of heterotopic ossification. *J Med Ultrasound*. 2014;22(4):222-227.
101. Liu XG, Zhang D. Evaluation of efficacy of corticosteroid and corticosteroid combined with botulinum toxin type A in the treatment of keloid and hypertrophic scars: a meta-analysis. *Aesthetic Plast Surg*. 2021;45(6):3037-3044.
102. Logsetty S, Shamlou A, Gawaziuk JP, et al. Mental health outcomes of burn: a longitudinal population-based study of adults hospitalized for burns. *Burns*. 2016;42(4):738-744.
103. Mason ST, Esselman P, Fraser R, et al. Return to work after burn injury: a systematic review. *J Burn Care Res*. 2012;33(1):101-109.
104. Matsen III FA. Compartmental syndrome. A unified concept. *Clin Orthop Relat Res*. 1975;113:8-14.
105. Medina A, Shankowsky H, Savaryn B, et al. Characterization of heterotopic ossification in burn patients. *J Burn Care Res*. 2014;35(3):251-256.
106. Miletta N, Siwy K, Hivnor C, et al. Fractional ablative laser therapy is an effective treatment for hypertrophic burn scars: a prospective study of objective and subjective outcomes. *Ann Surg*. 2021;274(6):e574-e580.
107. Momeni M, Hafezi F, Rahbar H, et al. Effects of silicone gel on burn scars. *Burns*. 2009;35(1):70-74.
108. Monsalve-Duarte S, Betancourt-Zapata W, Suarez-Canon N, et al. Music therapy and music medicine interventions with adult burn patients: a systematic review and meta-analysis. *Burns*. 2022;48(3):510-521.
109. Mony MP, Harmon KA, Hess R, et al. An updated review of hypertrophic scarring. *Cells*. 2023;12(5):678.
110. Morgan M, Deuis JR, Frosig-Jorgensen M, et al. Burn pain: a systematic and critical review of epidemiology, pathophysiology, and treatment. *Pain Med*. 2018;19(4):708-734.
111. Mustoe TA. Evolution of silicone therapy and mechanism of action in scar management. *Aesthetic Plast Surg*. 2008;32(1):82-92.
112. Nedelec B, Parry I, Acharya H, et al. Practice guidelines for cardiovascular fitness and strengthening exercise prescription after burn injury. *J Burn Care Res*. 2016;37(6):e539-e558.
113. Nedelec B, Serghiou MA, Niszcak J, et al. Practice guidelines for early ambulation of burn survivors after lower extremity grafts. *J Burn Care Res*. 2012;33(3):319-329.
114. Nunez JH, Clark AT. Burn patient metabolism and nutrition. *PM&R*. 2023;34(4):717-731.
115. O'Neil AM, Rush C, Griffard L, et al. Five-year retrospective analysis of a vented mobility algorithm in the burn ICU. *J Burn Care Res*. 2022;43(5):1129-1134.
116. Oosterwijk AM, Mouton LJ, Schouten H, et al. Prevalence of scar contractures after burn: a systematic review. *Burns*. 2017;43(1):41-49.
117. Orchard GR, Paratz JD, Blot S, et al. Risk factors in hospitalized patients with burn injuries for developing heterotopic ossification—a retrospective analysis. *J Burn Care Res*. 2015;36(4):465-470.
118. Palackic A, Suman OE, Porter C, et al. Rehabilitative exercise training for burn injury. *Sports Med*. 2021;51(12):2469-2482.
119. Pallua N, Kunsebeck HW, Noah EM. Psychosocial adjustments 5 years after burn injury. *Burns*. 2003;29(2):143-152.
120. Palmieri TL, Molitor F, Chan G, et al. Long-term functional outcomes in the elderly after burn injury. *J Burn Care Res*. 2012;33(4):497-503.
121. Palmu R, Suominen K, Vuola J, et al. Mental disorders after burn injury: a prospective study. *Burns*. 2011;37(4):601-609.
122. Pandya AA, Corkill HA, Goutos I. Sexual function following burn injuries: literature review. *J Burn Care Res*. 2015;36(6):e283-e293.
123. Parry IS, Schneider JC, Yelvington M, et al. Systematic review and expert consensus on the use of orthoses (splints and casts) with adults and children after burn injury to determine practice guidelines. *J Burn Care Res*. 2020;41(3):503-534.
124. Partridge J. About changing faces: promoting a good quality of life for people with visible disfigurements. *Burns*. 1997;23(2):186-187.
125. Patterson BW, Nguyen T, Pierre E, et al. Urea and protein metabolism in burned children: effect of dietary protein intake. *Metabolism*. 1997;46(5):573-578.
126. Patterson DR, Finch CP, Wiechman SA, et al. Premorbid mental health status of adult burn patients: comparison with a normative sample. *J Burn Care Rehabil*. 2003;24(5):347-350.
127. Paus R, Schmelz M, Biro T, et al. Frontiers in pruritus research: scratching the brain for more effective itch therapy. *J Clin Invest*. 2006;116(5):1174-1186.
128. Pavez RA, Martinez MP. Dysphagia in the burn patient: experience in a National Burn Reference Centre. *Burns*. 2019;45(5):1172-1181.
129. Pham TN, Carrouger GJ, Martinez E, et al. Predictors of discharge disposition in older adults with burns: a study of the Burn Model Systems. *J Burn Care Res*. 2015;36(6):607-612.
130. Pham TN, Goldstein R, Carrouger GJ, et al. The impact of discharge contracture on return to work after burn injury: a Burn Model System investigation. *Burns*. 2020;46(3):539-545.
131. Phoenix Society for Burn Survivors. *The Phoenix Society for Burn Survivors*. Accessed July 2, 2024. <http://phoenix-society.org>.
132. Porter C, Hardee JP, Herndon DN, et al. The role of exercise in the rehabilitation of patients with severe burns. *Exerc Sport Sci Rev*. 2015;43(1):34-40.
133. Porter C, Tompkins RG, Finnerty CC, et al. The metabolic stress response to burn trauma: current understanding and therapies. *Lancet*. 2016;388(10052):1417-1426.
134. Prasad A, Thode Jr HC, Sandoval S, et al. The association of patient and burn characteristics with itching and pain severity. *Burns*. 2019;45(2):348-353.
135. Prasad JK, Bowden ML, McDonald K, et al. Rehabilitation of burned patients with bilateral above-knee amputations. *Burns*. 1990;16(4):297-301.
136. Procter F. Rehabilitation of the burn patient. *Indian J Plast Surg*. 2010;43:S101-S113.
137. Pracek JT, Patterson DR, Heimbach DM. Inpatient depression in persons with burns. *J Burn Care Rehabil*. 2002;23(1):1-9.
138. Purohit M, Goldstein R, Nadler D, et al. Cognition in patients with burn injury in the inpatient rehabilitation population. *Arch Phys Med Rehabil*. 2014;95(7):1342-1349.
139. Qiao Z, Yang H, Jin L, et al. The efficacy and safety of botulinum toxin injections in preventing postoperative scars and improving scar quality: a systematic review and meta-analysis. *Aesthetic Plast Surg*. 2021;45(5):2350-2362.
140. Quinn T, Wasiak J, Cleland H. An examination of factors that affect return to work following burns: a systematic review of the literature. *Burns*. 2010;36(7):1021-1026.
141. Ramsey W, O'Neill C, Corona A. Burn excision within 48 hours portends better outcomes than standard management: a nationwide analysis. *J Trauma Acute Care Surg*. 2023;95(1):111-115.
142. Richards AM, Klaassen MF. Heterotopic ossification after severe burns: a report of three cases and review of the literature. *Burns*. 1997;23(1):64-68.
143. Rieck KM, Pagali S, Miller DM. Delirium in hospitalized older adults. *Hosp Pract (1995)*. 2020;48(1):S3-S16.
144. Rimaz S, Alavi CE, Sedighinejad A, et al. Effect of gabapentin on morphine consumption and pain after surgical debridement of

- burn wounds: a double-blind randomized clinical trial study. *Arch Trauma Res.* 2012;1(1):38-43.
145. Rimmer RB, Rutter CE, Lessard CR, et al. Burn care professionals' attitudes and practices regarding discussions of sexuality and intimacy with adult burn survivors. *J Burn Care Res.* 2010;31(4):579-589.
 146. Robinson E, Rumsey N, Partridge J. An evaluation of the impact of social interaction skills training for facially disfigured people. *Br J Plast Surg.* 1996;49(5):281-289.
 147. Rogers AD, Amaral A, Cartotto R, et al. Choosing wisely in burn care. *Burns.* 2022;48(5):1097-1103.
 148. Romanowski KS, Carson J, Pape K, et al. American Burn Association guidelines on the management of acute pain in the adult burn patient: a review of the literature, a compilation of expert opinion and next steps. *J Burn Care Res.* 2020;41(6):1152-1164.
 149. Rosenkranz KM, Sheridan R. Management of the burned trauma patient: balancing conflicting priorities. *Burns.* 2002;28(7):665-669.
 150. Roy TK, Uniyal A, Akhilesh, et al. Multifactorial pathways in burn injury-induced chronic pain: novel targets and their pharmacological modulation. *Mol Biol Rep.* 2022;49(12):12121-12132.
 151. Rumbach AF, Ward EC, Cornwell PL, et al. Incidence and predictive factors for dysphagia after thermal burn injury: a prospective cohort study. *J Burn Care Res.* 2011;32(6):608-616.
 152. Sabri A, Dabbous H, Dowli A, et al. The airway in inhalational injury: diagnosis and management. *Ann Burns Fire Disasters.* 2017;30(1):24-29.
 153. Sateia MJ, Buysse DJ, Krystal AD, et al. Clinical practice guideline for the pharmacologic treatment of chronic insomnia in adults: an American Academy of Sleep Medicine clinical practice guideline. *J Clin Sleep Med.* 2017;13(2):307-349.
 154. Schneider JC, Bassi S, Ryan CM. Employment outcomes after burn injury: a comparison of those burned at work and those burned outside of work. *J Burn Care Res.* 2011;32(2):294-301.
 155. Schneider JC, Harris NL, El Shami A, et al. A descriptive review of neuropathic-like pain after burn injury. *J Burn Care Res.* 2006;27(4):524-528.
 156. Schneider JC, Holavanahalli R, Helm P, et al. Contractures in burn injury: defining the problem. *J Burn Care Res.* 2006;27(4):508-514.
 157. Schneider JC, Qu HD. Neurologic and musculoskeletal complications of burn injuries. *PM&R.* 2011;22(2):261-275, vi.
 158. Schouten HJ, Nieuwenhuis MK, van Baar ME, et al. The prevalence and development of burn scar contractures: a prospective multicenter cohort study. *Burns.* 2019;45(4):783-790.
 159. Schweizer R, Pedrazzi N, Klein HJ, et al. Risk factors for mortality and prolonged hospitalization in electric burn injuries. *J Burn Care Res.* 2021;42(3):505-512.
 160. Shih JG, Shahrokhi S, Jeschke MG. Review of adult electrical burn injury outcomes worldwide: an analysis of low-voltage vs high-voltage electrical injury. *J Burn Care Res.* 2017;38(1):e293-e298.
 161. Shirakami E, Yamakawa S, Hayashida K. Strategies to prevent hypertrophic scar formation: a review of therapeutic interventions based on molecular evidence. *Burns Trauma.* 2020;8:tkz003.
 162. Shu F, Liu H, Lou X, et al. Analysis of the predictors of hypertrophic scarring pain and neuropathic pain after burn. *Burns.* 2022;48(6):1425-1434.
 163. Smith DL, Cairns BA, Ramadan F, et al. Effect of inhalation injury, burn size, and age on mortality: a study of 1447 consecutive burn patients. *J Trauma.* 1994;37(4):655-659.
 164. Sommerhalder C, Blears E, Murton AJ, et al. Current problems in burn hypermetabolism. *Curr Probl Surg.* 2020;57(1):100709.
 165. Soto CA, Albornoz CR, Pena V, et al. Prognostic factors for amputation in severe burn patients. *Burns.* 2013;39(1):126-129.
 166. Stewart BT, Sheckter CC, Nakarmi KK. Holistic approach to burn reconstruction and scar rehabilitation. *PM&R.* 2023;34(4):883-904.
 167. Stover SL, Hataway CJ, Zeiger HE. Heterotopic ossification in spinal cord-injured patients. *Arch Phys Med Rehabil.* 1975;56(5):199-204.
 168. Summer GJ, Puntillo KA, Miaskowski C, et al. Burn injury pain: the continuing challenge. *J Pain.* 2007;8(7):533-548.
 169. Tan WH, Goldstein R, Gerrard P, et al. Outcomes and predictors in burn rehabilitation. *J Burn Care Res.* 2012;33(1):110-117.
 170. Tawfik AA, Ali RA. Evaluation of botulinum toxin type A for treating post burn hypertrophic scars and keloid in children: an intra-patient randomized controlled study. *J Cosmet Dermatol.* 2023;22(4):1256-1260.
 171. Thananopavarn P, Hill III JJ. Rehabilitation of the complex burn patient with multiple injuries or comorbidities. *Clin Plast Surg.* 2017;44(4):695-701.
 172. Thombs BD, Notes LD, Lawrence JW, et al. From survival to socialization: a longitudinal study of body image in survivors of severe burn injury. *J Psychosom Res.* 2008;64(2):205-212.
 173. Tiwari VK. Burn wound: how it differs from other wounds? *Indian J Plast Surg.* 2012;45(2):364-373.
 174. Tu Y, Lineaweaver WC, Zheng X, et al. Burn-related peripheral neuropathy: a systematic review. *Burns.* 2017;43(4):693-699.
 175. Turner E, Robinson DM, Roaten K. Psychological issues. *PM&R.* 2023;34(4):849-866.
 176. Ullrich PM, Askay SW, Patterson DR. Pain, depression, and physical functioning following burn injury. *Rehabil Psychol.* 2009;54(2):211-216.
 177. Van Loey NE, Van Son MJ. Psychopathology and psychological problems in patients with burn scars: epidemiology and management. *Am J Clin Dermatol.* 2003;4(4):245-272.
 178. Van Loey NEE, de Jong AEE, Hofland HWC, et al. Role of burn severity and posttraumatic stress symptoms in the co-occurrence of itch and neuropathic pain after burns: a longitudinal study. *Front Med (Lausanne).* 2022;9:997183.
 179. Veeravagu A, Yoon BC, Jiang B, et al. National trends in burn and inhalation injury in burn patients: results of analysis of the nationwide inpatient sample database. *J Burn Care Res.* 2015;36(2):258-265.
 180. Villarreal E, Wolf SE, Golovko G, et al. Opioid prescription and opioid disorders in burns: a large database analysis from 1990 to 2019. *Burns.* 2023;49(8):1845-1853.
 181. Wang Y, Wang J, Zhang J, et al. Effectiveness and safety of botulinum toxin type A injection for scar prevention: a systematic review and meta-analysis. *Aesthetic Plast Surg.* 2019;43(5):1241-1249.
 182. Wardhan R, Fahy BG. Regional anesthesia and acute pain management for adult patients with burns. *J Burn Care Res.* 2023;44(4):791-799.
 183. Wibbenmeyer L, Eid A, Liao J, et al. Gabapentin is ineffective as an analgesic adjunct in the immediate postburn period. *J Burn Care Res.* 2014;35(2):136-142.
 184. Wiechman SA, Ptacek JT, Patterson DR, et al. Rates, trends, and severity of depression after burn injuries. *J Burn Care Rehabil.* 2001;22(6):417-424.
 185. Williams FN, Herndon DN, Jeschke MG. The hypermetabolic response to burn injury and interventions to modify this response. *Clin Plast Surg.* 2009;36(4):583-596.
 186. Won YH, Cho YS, Joo SY, et al. Respiratory characteristics in patients with major burn injury and smoke inhalation. *J Burn Care Res.* 2022;43(1):70-76.
 187. Wong L, Turner L. Treatment of post-burn neuropathic pain: evaluation of pregablin. *Burns.* 2010;36(6):769-772.
 188. Wright M, Lee JA. Multimodal analgesia and discharge opioid requirements in burn patients. *J Burn Care Res.* 2020;41(5):963-966.
 189. Wrigley M, Trotman BK, Dimick A, et al. Factors relating to return to work after burn injury. *J Burn Care Rehabil.* 1995;16(4):445-450, discussion 444.
 190. Yowler CJ, Mazingo DW, Ryan JB, et al. Factors contributing to delayed extremity amputation in burn patients. *J Trauma.* 1998;45(3):522-526.
 191. Yu YM, Tompkins RG, Ryan CM, et al. The metabolic basis of the increase of the increase in energy expenditure in severely burned patients. *JPEN.* 1999;23(3):160-168.
 192. Zhuang Z, Li Y, Wei X. The safety and efficacy of intralesional triamcinolone acetonide for keloids and hypertrophic scars: a systematic review and meta-analysis. *Burns.* 2021;47(5):987-998.